

RDTU

Supplementary Materials

Does LLM-based Time Series Forecasting Really Need Pre-Alignment?

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<https://github.com/LiRin/RDTU-Supplementary>

This companion restores the expanded explanations, experiments, figures and appendices omitted or condensed in the ICASSP manuscript. Sections 1–5 provide context and expanded analyses. Appendices A–G contain training details, complete forecasting results, statistical analyses, generalization experiments, the sample-mining algorithm, qualitative examples, prompts and societal impacts.

Numerical table entries are transcribed from the supplied full manuscript. Wide result tables are reorganized by dataset for readability; the reported averages and missing-value markers are preserved. Source-reported ranking counts are distinguished from newly computed quantities. The ETT transfer summary follows the final manuscript’s correction: lowest average MSE in seven of eight directions.

Contents

1	Introduction	3
2	Related Works	4
3	Methodology	4
3.1	Overview	4
3.2	Supervised Temporal Instruction Tuning	4
3.3	Reinforcement-Driven Forecasting Refinement	4
3.3.1	RDFR Data Construction Strategy	5
3.3.2	Temporal Refinement Reward System	6
4	Main Results	7
4.1	Long-term Forecasting	8
4.2	Short-term Forecasting	9
4.3	Few-shot Forecasting	11
4.4	Zero-shot Forecasting	11
4.5	Showcases Analysis	12
4.6	Ablation and Sensitivity Analysis	12
5	Conclusion and Limitations	14
A	Experimental Details	14
A.1	Implementation	14
A.2	Evaluation Metrics	14
A.3	Dataset Details	15
A.4	Statistical Significance Analysis	15
B	Long-term and Short-term Forecasting	17
B.1	Long-term Forecasting	17
B.2	Short-term Forecasting	21

C Few-Shot and Zero-Shot Forecasting	23
C.1 Few-Shot Forecasting	23
C.2 Zero-Shot Forecasting	28
C.3 Zero-shot Length Generalization	30
C.4 Cross-Domain Zero-Shot Generalization	30
D Priority-Based Dual-Stream Hard Sample Mining	31
E Showcases	32
F Prompt	33
G Societal Impacts	34
Source and formatting notes	34

1 Introduction

Time series forecasting is pivotal for modern decision-making, enabling strategic resource allocation in critical sectors ranging from energy [31] and traffic [10] to finance and weather [7]. Driven by this practical value, the field has rapidly evolved from traditional statistical methods to sophisticated deep learning paradigms [17, 23]. Consequently, developing robust models capable of generalizing across these diverse, dynamic scenarios remains a critical challenge in machine learning. In pursuit of superior forecasting, current research primarily diverges into two paths. Specialized Transformer architectures, such as PatchTST [19] and iTransformer [16], effectively capture temporal dependencies but often lack the cross-domain transferability of foundation models. Consequently, recent works have adapted Large Language Models (LLMs) [8, 33] and Large Vision Models (LVMs) [21, 30] for time series tasks. However, the prevailing “Align-then-Tune” paradigm, exemplified by Time-LLM [8], treats time series as a foreign modality. This approach necessitates complex **pre-alignment** modules such as reprogramming layers to map continuous numerical tokens into the semantic space prior to fine-tuning. However, “Align-then-Tune” strategies face two intrinsic limitations. First, a fundamental **granularity mismatch** persists between continuous high-dimensional time series and discrete linguistic space; forcing numerical fluctuations into embeddings inevitably causes *information loss*, obscuring critical temporal dynamics during the translation process. Second, reliance on explicit pre-alignment creates significant **architectural overhead**. These auxiliary modules act as bottlenecks, effectively preventing the LLM from directly applying its inherent reasoning capabilities to the raw numerical data.

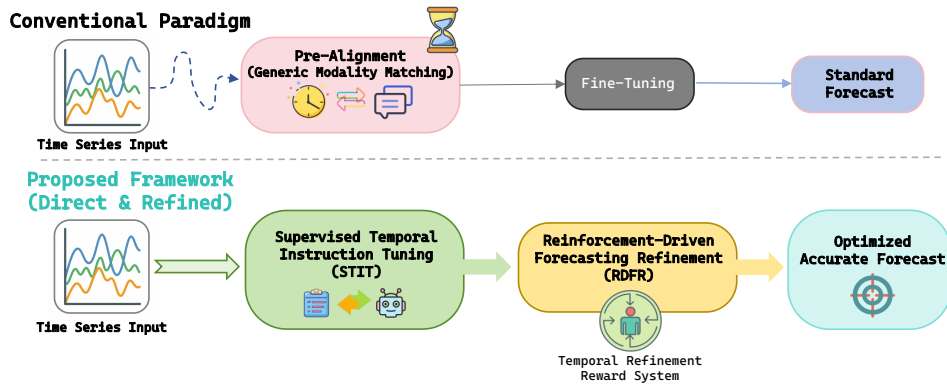


Figure S1: Comparison of forecasting paradigms. Conventional approaches [8, 33] rely on pre-alignment modules with limitations of granularity mismatch and architectural overhead to bridge modality gaps, when our proposed framework **RDTU** adopts a direct unification strategy, bypassing pre-alignment via **Supervised Temporal Instruction Tuning (STIT)** and strictly enforcing precision through **Reinforcement-Driven Forecasting Refinement (RDFR)**.

To overcome alignment limitations, we explore a paradigm that views forecasting as a native token generation task, treating numerical values directly as textual strings within unified prompts to bypass external encoders projecting continuous time series data into a high-dimensional semantic space. However, this direct approach faces two critical hurdles: **precision misalignment**, where standard causal modeling prioritizes semantic plausibility over the rigorous numerical accuracy required for regression, and **structural hallucinations**, where off-the-shelf models frequently fail to adhere to strict output formats or sequence lengths, rendering predictions unusable for downstream applications.

To address these challenges and fully leverage the native reasoning of LLMs, we propose the **Reinforced Direct Temporal Unification (RDTU)** framework. RDTU bridges the gap between discrete textual generation and continuous numerical forecasting through a progressive pipeline. Specifically, we first introduce **Supervised Temporal Instruction Tuning (STIT)** to align the model with the structure of time-series prompts. Crucially, to tackle the precision and formatting challenges inherent in textual token generation, we design a **Reinforcement-Driven Forecasting Refinement (RDFR)** mechanism. This stage employs a specialized reward system to rigorously enforce numerical accuracy and syntactic validity, ensuring the LLM acts as a precise forecaster. The following empirical evaluations confirm that this streamlined approach yields State-of-the-Art results while simplifying the training process. Our main contributions are summarized as follows:

- We propose **STIT**, a direct unification paradigm that bypasses auxiliary pre-alignment by treating time series as native text which not only simplifies training complexity but also achieves robust instruction adherence and comparable forecasting performance.
- We introduce **RDFR**, a specialized reinforcement learning system utilizing dual-stream hard sample mining and multi-objective rewards which bridges the gap between discrete generation and continuous precision.

- We validate the efficacy of our progressive **RDTU** framework through extensive experiments. Results demonstrate that this synergy yields superior accuracy and generalization.

2 Related Works

RDTU studies direct numerical text generation with supervised instruction tuning and reinforcement-driven refinement for time series forecasting. In the following, we review existing literature across two key dimensions: traditional Transformer-based approaches and emerging LLM-based methodologies for time series forecasting (TSF).

Transformer-based TSF Deep learning architectures have fundamentally reshaped time series forecasting, with Transformers initially gaining prominence due to their sequence modeling capabilities [23]. To address computational bottlenecks, efficient variants such as Informer [31], Reformer [9], and Pyraformer [12] introduced sparse attention mechanisms to reduce complexity for long-sequence modeling. Subsequently, decomposition-based architectures like Autoformer [26] and FEDformer [32] enhanced stability on non-stationary data by explicitly disentangling seasonal and trend components. Most recently, PatchTST [19] and iTransformer [16] have achieved state-of-the-art results by employing patching techniques and inverted embeddings to effectively capture multivariate correlations and local semantic contexts.

LLM-based TSF The unprecedented capabilities of foundation models have catalyzed cross-modality learning in time series, evolving from the zero-shot textual generation of LLMTime [18] to alignment-based adaptations. Notably, GPT4TS [33] validates freezing LLM backbones with fine-tuned embeddings, while Time-LLM [8] employs a reprogramming framework to align temporal patches with text prototypes, and UniTime [14] utilizes textual instructions for cross-domain unification. Concurrently, research has expanded into visual modalities: VisionTS [3] adapts masked autoencoders for zero-shot forecasting by treating time series as visual signals; Time-VLM [30] leverages vision-language models by converting data into visual plots; and DMMV-A [21] enhances large vision models through multi-view constraints to optimize long-term forecasting stability. Beyond adapting general-purpose LLMs, recent studies have also explored dedicated time-series foundation models pre-trained on large-scale temporal corpora, such as Chronos-2, TimesFM, Moirai, and MOMENT [1, 4, 6, 25].

3 Methodology

3.1 Overview

We propose **Reinforced Direct Temporal Unification (RDTU)**, a framework that exploits the intrinsic reasoning ability of Large Language Models (LLMs) for time series forecasting without complex modality alignment modules. As shown in Figure S2, RDTU follows a progressive two-stage training pipeline. First, **Supervised Temporal Instruction Tuning (STIT)** builds structure-aware instruction data to endow the model with basic forecasting ability and domain-specific instruction following, as detailed in subsection 3.2. Second, **Reinforcement-Driven Forecasting Refinement (RDFR)** further improves regression precision through task-specific rewards that enforce output format and refine prediction accuracy, as described in subsection 3.3.

3.2 Supervised Temporal Instruction Tuning

To establish foundational capabilities for time-series reasoning and strict instruction adherence, we initiate the pipeline with **Supervised Temporal Instruction Tuning (STIT)**. Diverging from methods that rely on auxiliary encoders, we employ a direct serialization strategy where historical numerical data is transformed into discrete text strings. These sequences are seamlessly integrated into a structured natural language prompt composed of four semantic elements: an expert persona definition, domain-specific metadata, the serialized historical sequence, and the specific prediction directive. This unified textual format allows the pre-trained LLM backbone to be fine-tuned directly via standard causal language modeling, as detailed in Appendix F.

3.3 Reinforcement-Driven Forecasting Refinement

While STIT provides a good initialization, the model often struggles with strict numerical output regarding the formats and long-horizon precision. To address these residual issues, we introduce the **RDFR** system, which is composed of two core contributions: a *Priority-Based Dual-Stream Hard Sample Mining* strategy for constructing

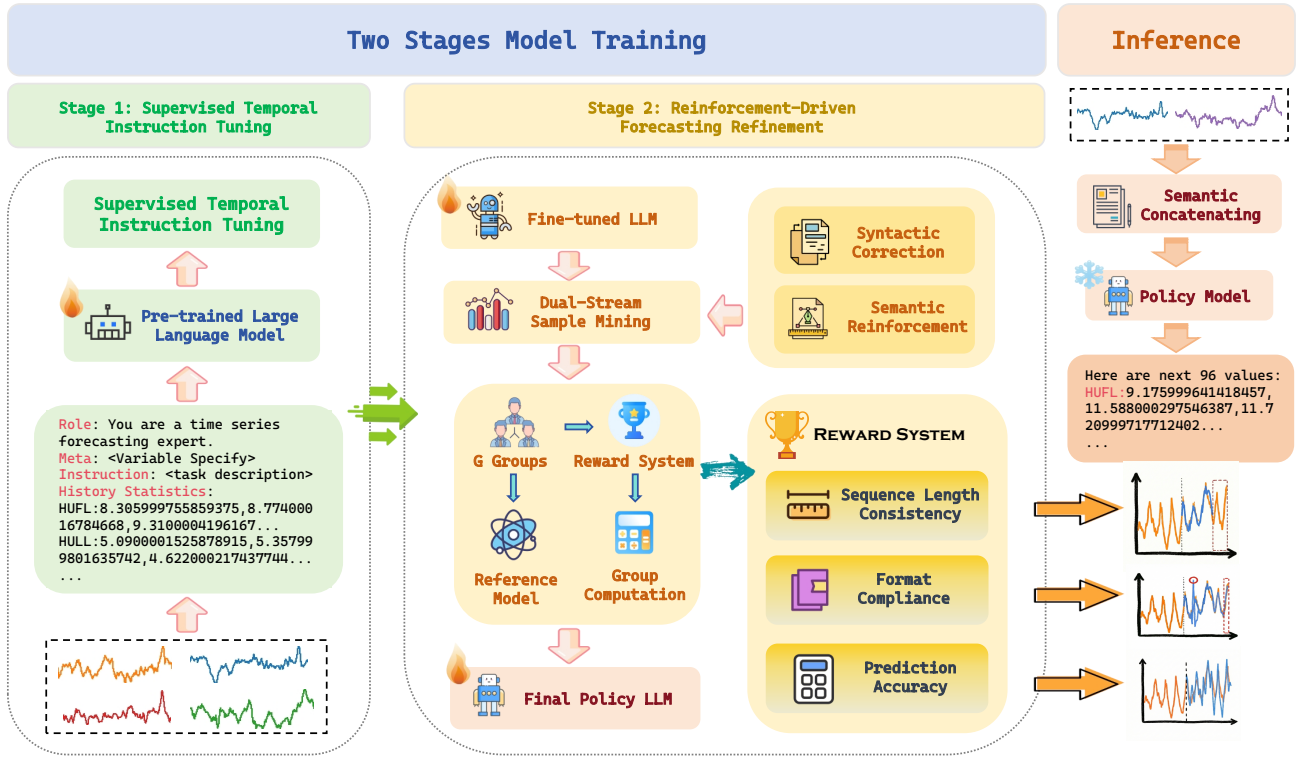


Figure S2: The overall architecture of the RDTU framework. The pipeline proceeds in two stages: (1) **Supervised Temporal Instruction Tuning (STIT)** initializes the model with foundational forecasting capabilities via structured prompts; (2) **Reinforcement-Driven Forecasting Refinement (RDFR)** employs a Reinforcement Learning system to enforce numerical precision and formatting constraints.

high-leverage RDFR training data, and a *Temporal Refinement Reward System* designed to guide the optimization process.

3.3.1 RDFR Data Construction Strategy

To address the instruction-following failures and precision deficits that persist after the STIT stage, we propose a **Priority-Based Dual-Stream Hard Sample Mining** strategy. As illustrated in Figure S3, we decouple the data selection into two parallel streams, Syntactic Correction ($\mathcal{D}_{syn}$) and Semantic Reinforcement ($\mathcal{D}_{sem}$), which are unified to form the final Reinforcement-Driven Forecasting Refinement Dataset $\mathcal{D}_{RDFR} = \mathcal{D}_{syn} \cup \mathcal{D}_{sem}$. The complete algorithm of data construction strategy is shown in Appendix D.

Syntactic Correction Stream. The primary motivation of this stream is to eliminate "formatting hallucinations," such as non-numerical outputs or sequence length inconsistencies, which render predictions invalid. To quantify structural integrity, we define a format completeness score S_{fmt} :

$$S_{fmt}^{(i)} = \mathbb{I}_{\text{valid}}(\hat{y}_i) \times \left(1 - \min \left(1, \frac{|L_{pred}^{(i)} - L_{target}|}{L_{target}} \right) \right) \quad (\text{S1})$$

where $\hat{y}_i$ is the predicted sequence, and $\mathbb{I}_{\text{valid}}$ is a binary indicator that returns 0 if forbidden patterns are detected. The term involving $L_{pred}^{(i)}$ and L_{target} penalizes deviations from the ground-truth sequence length. We construct $\mathcal{D}_{syn}$ by prioritizing samples with the lowest S_{fmt} scores, effectively forcing the model to confront its most severe structural failures.

To construct $\mathcal{D}_{syn}$, the algorithm does not rely solely on ranking samples by their format scores (S_{fmt}); it also actively detects and prioritizes specific failure patterns (e.g., missing decimal points) to enhance the model's targeted correction capabilities against severe structural errors.

For the complete algorithmic procedure and implementation details, please refer to Appendix D.

Semantic Reinforcement Stream. To refine regression accuracy without overfitting to outliers, this stream targets "hard" yet representative samples. We employ a *Density-Aware Stratified Sampling* strategy on the error

manifold. Focusing on samples where the error exceeds the population mean ($L_{mse}^{(i)} > \mu$), we partition the error distribution into K bins and assign sampling quotas proportional to density:

$$n_k = N_{sem} \times \frac{|\text{Bin}_k|}{\sum_{j=1}^K |\text{Bin}_j|} \quad (\text{S2})$$

Here, n_k represents the number of samples drawn from the k -th error interval based on the total semantic budget N_{sem} , and $|\text{Bin}_k|$ denotes the sample count in that interval. This ensures $\mathcal{D}_{sem}$ captures the diverse modes of numerical difficulty inherent in the STIT model.

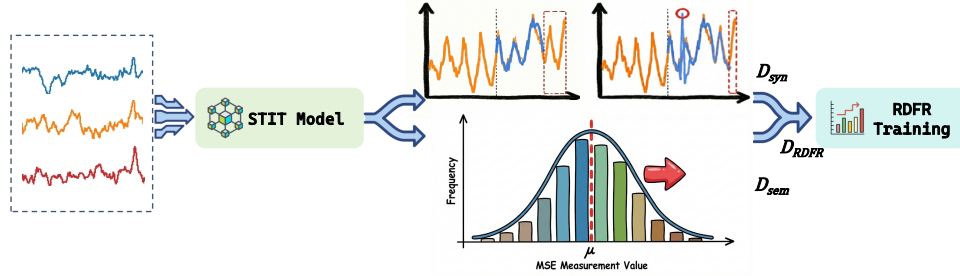


Figure S3: Priority-Based Dual-Stream Hard Sample Mining strategy

3.3.2 Temporal Refinement Reward System

To align the model's policy with the standards of time-series forecasting, we employ Group Relative Policy Optimization (GRPO) [5] as our Reinforcement Forecasting Refinement framework. Therefore, we design a forecasting refinement reward function R_{total} comprising three distinct components, each targeting a specific failure mode observed in STIT models.

Sequence Length Consistency Reward (R_{len}). To mitigate "counting failures" where models drift from the required horizon, we replace standard linear penalties, which often lack sensitivity to "off-by-one" errors, with a non-linear exponential penalty. This acts as a soft constraint, imposing sharp penalties for minimal deviations to enforce rapid convergence to the target length L_{target} :

$$R_{len} = \exp(-\alpha \cdot |L_{pred} - L_{target}|) \quad (\text{S3})$$

where α controls reward sensitivity, ensuring exact length adherence for high scores.

Format Compliance Reward (R_{fmt}). To overcome the "sparse signal" problem where binary validity checks fail to distinguish between total hallucination and minor formatting errors, we employ *reward shaping*. We decompose structural compliance into progressive sub-goals to provide dense gradient feedback:

$$R_{fmt} = \frac{1}{N} \sum_{i=1}^N \left(w_1 \cdot \mathbb{I}(\text{is_number}_i) + w_2 \cdot \mathbb{I}(\text{is_float}_i) + w_3 \cdot \mathbb{I}(\text{precision}_i) \right) \quad (\text{S4})$$

where N represents the total number of predicted numerical values, and the indicator functions $\mathbb{I}(\cdot)$ verify token-level adherence to numeric character usage, decimal presence, and strict decimal precision as specified by the task, respectively, weighted by $w_{\{1,2,3\}}$.

Prediction Accuracy Reward (R_{acc}). Since unbounded MSE loss can destabilize reward distributions and lacks sensitivity in high-precision regimes, we map the error to a bounded $(0, 1]$ interval via exponential decay. This ensures distinct feedback even as the model approaches saturation:

$$R_{acc} = \exp(-\beta \cdot \text{MSE}(Y_{pred}, Y_{gt})) \quad (\text{S5})$$

where β regulates the strictness of the accuracy requirement, preventing outlier dominance while maintaining significant gradient magnitude for small improvements.

The final reward is computed as a weighted sum $R_{total} = \lambda_{len} R_{len} + \lambda_{fmt} R_{fmt} + \lambda_{acc} R_{acc}$, balancing structural validity with regression precision.

4 Main Results

Table S1: Long-term forecasting: horizon averages. Each entry is MSE / MAE; lower values are better.

ETTh1		ETTh2	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.399 / 0.413	RDTU	0.337 / 0.377
GPT4TS	0.418 / 0.421	GPT4TS	0.354 / 0.389
Time-LLM	0.418 / 0.432	Time-LLM	0.361 / 0.396
DMMV-A	0.395 / 0.414	DMMV-A	0.337 / 0.388
Time-VLM	0.405 / 0.420	Time-VLM	0.341 / 0.391
VisionTS	0.407 / 0.419	VisionTS	0.351 / 0.386
PatchTST	0.413 / 0.431	PatchTST	0.330 / 0.379
CycleNet	0.415 / 0.426	CycleNet	0.355 / 0.398
TimesNet	0.458 / 0.450	TimesNet	0.414 / 0.427
DLinear	0.423 / 0.430	DLinear	0.431 / 0.447
FEDformer	0.440 / 0.460	FEDformer	0.437 / 0.449

Table S1 (continued).

ETTm1		ETTm2	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.340 / 0.367	RDTU	0.250 / 0.309
GPT4TS	0.363 / 0.378	GPT4TS	0.254 / 0.311
Time-LLM	0.356 / 0.377	Time-LLM	0.261 / 0.316
DMMV-A	0.340 / 0.371	DMMV-A	0.256 / 0.317
Time-VLM	0.351 / 0.376	Time-VLM	0.248 / 0.311
VisionTS	0.344 / 0.373	VisionTS	0.267 / 0.327
PatchTST	0.351 / 0.381	PatchTST	0.255 / 0.315
CycleNet	0.355 / 0.379	CycleNet	0.251 / 0.309
TimesNet	0.400 / 0.406	TimesNet	0.291 / 0.333
DLinear	0.357 / 0.379	DLinear	0.267 / 0.332
FEDformer	0.448 / 0.452	FEDformer	0.305 / 0.349

Table S1 (continued).

Illness		Electricity	
Model	MSE / MAE	Model	MSE / MAE
RDTU	1.559 / 0.812	RDTU	0.156 / 0.246
GPT4TS	1.871 / 0.852	GPT4TS	0.170 / 0.263
Time-LLM	2.018 / 0.894	Time-LLM	0.165 / 0.259
DMMV-A	1.407 / 0.771	DMMV-A	0.158 / 0.248
Time-VLM	–	Time-VLM	0.172 / 0.272
VisionTS	1.482 / 0.796	VisionTS	0.159 / 0.250
PatchTST	1.443 / 0.798	PatchTST	0.162 / 0.253
CycleNet	2.187 / 0.992	CycleNet	0.158 / 0.250
TimesNet	2.139 / 0.931	TimesNet	0.193 / 0.295
DLinear	2.169 / 1.041	DLinear	0.166 / 0.264
FEDformer	2.847 / 1.144	FEDformer	0.214 / 0.327

Table S1 (continued).

Weather		Traffic	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.221 / 0.251	RDTU	0.389 / 0.255
GPT4TS	0.227 / 0.255	GPT4TS	0.421 / 0.274
Time-LLM	0.244 / 0.270	Time-LLM	0.422 / 0.281
DMMV-A	0.217 / 0.256	DMMV-A	0.389 / 0.257
Time-VLM	0.224 / 0.263	Time-VLM	0.419 / 0.304
VisionTS	0.225 / 0.258	VisionTS	0.386 / 0.256
PatchTST	0.226 / 0.264	PatchTST	0.391 / 0.264
CycleNet	0.242 / 0.278	CycleNet	0.421 / 0.289
TimesNet	0.259 / 0.287	TimesNet	0.620 / 0.336
DLinear	0.249 / 0.300	DLinear	0.434 / 0.295
FEDformer	0.309 / 0.360	FEDformer	0.610 / 0.376

Source-reported ranking counts. RDTU: 9, GPT4TS: 0, Time-LLM: 0, DMMV-A: 5, Time-VLM: 1, VisionTS: 1, PatchTST: 1, CycleNet: 1, TimesNet: 0, DLinear: 0, FEDformer: 0. The original counting convention is not specified; these values are retained without recalculation.

Table S2: M4 forecasting: weighted averages. Lower values are better.

Average			
Model	SMAPE	MASE	OWA
RDTU	11.979	1.599	0.859
Time-LLM	11.983	1.595	0.859
GPT4TS	12.69	1.808	0.94
TimesNet	12.88	1.836	0.955
PatchTST	12.059	1.623	0.869
N-HiTS	12.035	1.625	0.869
N-BEATS	12.25	1.698	0.896
ETSformer	14.718	2.408	1.172
LightTS	13.525	2.111	1.051
DLinear	13.639	2.095	1.051
FEDformer	13.16	1.775	0.949
Stationary	12.780	1.756	0.930
Autoformer	12.909	1.771	0.939
Informer	14.086	2.718	1.230
Reformer	18.200	4.223	1.775

Datasets. We adopt 8 widely used MTS benchmarks: ETT (Electricity Transformer Temperature) [31], including ETTh1, ETTh2, ETTm1, ETTm2; Weather [26], Illness [26], Traffic [26], and Electricity [22], which have been extensively adopted for benchmarking long-term forecasting models [27]. The prediction horizon H is set to $\{24, 36, 48, 60\}$ for Illness, and $\{96, 192, 336, 720\}$ for the remaining datasets.

Baselines. We compare RDTU with SOTA methods, categorized as follows: (1) **VLM-based models:** Time-VLM [30]; (2) **LVM-based models:** VisionTS [3] and DMMV-A [21]; (3) **LLM-based models:** Time-LLM [8] and GPT4TS [33]; (4) **Transformer-based models:** PatchTST [19], FEDformer [32], Autoformer [26], Stationary [15], ETSformer [24], and Informer [31]; (5) **Non-Transformer models:** DLinear [29], TimesNet [27], and CycleNet [11]. In short-term forecasting, we compare our model with N-HiTS [2] and N-BEATS [20].

Implementation. We implement our framework using **Qwen2.5-7B-Instruct** [28] as the primary backbone. All experiments are conducted on 8 **NVIDIA A100 (80GB)** GPUs. More details such as specific training expenses are shown in [Appendix A](#).

4.1 Long-term Forecasting

Setups. We conduct extensive evaluations on eight widely-used real-world benchmarks: ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity (ECL), Traffic, and ILI.

Results. [Table S1](#) summarizes the long-term forecasting performance. **RDTU** achieves superior accuracy, securing the highest number of best results (**9 wins**) across all benchmarks. It achieves highly competitive state-of-the-art-level performance, while remaining competitive with the strongest specialized numerical and

vision-based baselines. Notably, RDTU establishes a significant lead over the strongest language-based baselines, reducing MSE by approximately **5.5%** compared to Time-LLM on the *Electricity* dataset and **4.5%** versus GPT4TS on *ETTh1*. These results confirm that our direct unification strategy, devoid of alignment bottlenecks, captures complex temporal dependencies with greater precision than competing paradigms.

4.2 Short-term Forecasting

Setups. We extend our evaluation to short-term forecasting using the M4 benchmark [17].

We employ Symmetric Mean Absolute Percentage Error (SMAPE), Mean Absolute Scaled Error (MASE), and Overall Weighted Average (OWA) as the evaluation metrics.

Results. As presented in Table S2, RDTU demonstrates robust short-horizon performance, achieving the lowest SMAPE of **11.979** and a state-of-the-art OWA of **0.859**. It significantly outperforms baselines like GPT4TS (OWA 0.940) and TimesNet (OWA 0.955), proving that our skip-alignment strategy effectively preserves local temporal details. Furthermore, RDTU surpasses specialized models such as N-HiTS and PatchTST (both OWA 0.869). While matching Time-LLM’s competitive performance, RDTU achieves these results via a more streamlined direct unification architecture, validating the efficacy of the RDFR system in optimizing precision across varying temporal scales.

Table S3: Few-shot forecasting: 10% horizon averages. Each entry is MSE / MAE; lower values are better.

ETTh1		ETTh2	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.551 / 0.513	RDTU	0.367 / 0.389
Time-LLM	0.556 / 0.522	Time-LLM	0.370 / 0.394
GPT4TS	0.590 / 0.525	GPT4TS	0.397 / 0.421
DLinear	0.691 / 0.600	DLinear	0.605 / 0.538
PatchTST	0.633 / 0.542	PatchTST	0.415 / 0.431
TimesNet	0.869 / 0.628	TimesNet	0.479 / 0.465
FEDformer	0.639 / 0.561	FEDformer	0.466 / 0.475
Autoformer	0.702 / 0.596	Autoformer	0.488 / 0.499
Stationary	0.915 / 0.639	Stationary	0.462 / 0.455
ETSformer	1.180 / 0.834	ETSformer	0.894 / 0.713
LightTS	1.375 / 0.877	LightTS	2.655 / 1.160
Informer	1.199 / 0.809	Informer	3.872 / 1.513
Reformer	1.249 / 0.833	Reformer	3.485 / 1.486

Table S3 (continued).

ETTm1		ETTm2	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.402 / 0.424	RDTU	0.274 / 0.319
Time-LLM	0.404 / 0.427	Time-LLM	0.277 / 0.323
GPT4TS	0.464 / 0.441	GPT4TS	0.293 / 0.335
DLinear	0.411 / 0.429	DLinear	0.316 / 0.368
PatchTST	0.501 / 0.466	PatchTST	0.296 / 0.343
TimesNet	0.677 / 0.537	TimesNet	0.320 / 0.353
FEDformer	0.722 / 0.605	FEDformer	0.463 / 0.488
Autoformer	0.802 / 0.628	Autoformer	1.342 / 0.930
Stationary	0.797 / 0.578	Stationary	0.332 / 0.366
ETSformer	0.980 / 0.714	ETSformer	0.447 / 0.487
LightTS	0.971 / 0.705	LightTS	0.987 / 0.756
Informer	1.192 / 0.821	Informer	3.370 / 1.440
Reformer	1.426 / 0.856	Reformer	3.978 / 1.587

Table S3 (continued).

Weather		Electricity	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.232 / 0.270	RDTU	0.173 / 0.268
Time-LLM	0.234 / 0.273	Time-LLM	0.175 / 0.270
GPT4TS	0.238 / 0.275	GPT4TS	0.176 / 0.269
DLinear	0.241 / 0.283	DLinear	0.180 / 0.280
PatchTST	0.242 / 0.279	PatchTST	0.180 / 0.273
TimesNet	0.279 / 0.301	TimesNet	0.323 / 0.392
FEDformer	0.284 / 0.324	FEDformer	0.346 / 0.427
Autoformer	0.300 / 0.342	Autoformer	0.431 / 0.478
Stationary	0.318 / 0.323	Stationary	0.444 / 0.480
ETSformer	0.318 / 0.360	ETSformer	0.660 / 0.617
LightTS	0.289 / 0.322	LightTS	0.441 / 0.489
Informer	0.597 / 0.495	Informer	1.195 / 0.891
Reformer	0.546 / 0.469	Reformer	0.965 / 0.768

Table S3 (continued).

Traffic	
Model	MSE / MAE
RDTU	0.426 / 0.303
Time-LLM	0.429 / 0.306
GPT4TS	0.440 / 0.310
DLinear	0.447 / 0.313
PatchTST	0.430 / 0.305
TimesNet	0.951 / 0.535
FEDformer	0.663 / 0.425
Autoformer	0.749 / 0.446
Stationary	1.453 / 0.815
ETSformer	1.914 / 0.936
LightTS	1.248 / 0.684
Informer	1.534 / 0.811
Reformer	1.551 / 0.821

Source-reported ranking counts. RDTU: 14, Time-LLM: 0, GPT4TS: 0, DLinear: 0, PatchTST: 0, TimesNet: 0, FEDformer: 0, Autoformer: 0, Stationary: 0, ETSformer: 0, LightTS: 0, Informer: 0, Reformer: 0. The original counting convention is not specified; these values are retained without recalculation.

Table S4: Zero-shot ETT transfer: horizon averages. Each entry is MSE / MAE; lower values are better.

ETTh1 → ETTh2		ETTh1 → ETTm2	
Model	MSE / MAE	Model	MSE / MAE
RDTU	0.344 / 0.379	RDTU	0.266 / 0.333
Time-LLM	0.353 / 0.387	Time-LLM	0.273 / 0.340
LLMTime	0.992 / 0.708	LLMTime	1.867 / 0.869
GPT4TS	0.406 / 0.422	GPT4TS	0.325 / 0.363
DLinear	0.493 / 0.488	DLinear	0.415 / 0.452
PatchTST	0.380 / 0.405	PatchTST	0.314 / 0.360
TimesNet	0.421 / 0.431	TimesNet	0.327 / 0.361
Autoformer	0.582 / 0.548	Autoformer	0.457 / 0.483

Table S4 (continued).

ETTh2 $\rightarrow$ ETTh1	
Model	MSE / MAE
RDTU	0.480 / 0.471
Time-LLM	0.479 / 0.474
LLMTime	1.961 / 0.981
GPT4TS	0.757 / 0.578
DLinear	0.703 / 0.574
PatchTST	0.565 / 0.513
TimesNet	0.865 / 0.621
Autoformer	0.757 / 0.608

ETTh2 $\rightarrow$ ETTm2	
Model	MSE / MAE
RDTU	0.268 / 0.329
Time-LLM	0.272 / 0.341
LLMTime	1.867 / 0.869
GPT4TS	0.335 / 0.370
DLinear	0.328 / 0.386
PatchTST	0.325 / 0.365
TimesNet	0.342 / 0.376
Autoformer	0.366 / 0.411

Table S4 (continued).

ETTh1 $\rightarrow$ ETTh2	
Model	MSE / MAE
RDTU	0.378 / 0.408
Time-LLM	0.381 / 0.412
LLMTime	0.992 / 0.708
GPT4TS	0.433 / 0.439
DLinear	0.464 / 0.475
PatchTST	0.439 / 0.438
TimesNet	0.457 / 0.454
Autoformer	0.470 / 0.479

ETTh1 $\rightarrow$ ETTm2	
Model	MSE / MAE
RDTU	0.261 / 0.316
Time-LLM	0.268 / 0.320
LLMTime	1.867 / 0.869
GPT4TS	0.313 / 0.348
DLinear	0.335 / 0.389
PatchTST	0.296 / 0.334
TimesNet	0.322 / 0.354
Autoformer	0.469 / 0.484

Table S4 (continued).

ETTh2 $\rightarrow$ ETTm2	
Model	MSE / MAE
RDTU	0.351 / 0.395
Time-LLM	0.354 / 0.400
LLMTime	0.992 / 0.708
GPT4TS	0.435 / 0.443
DLinear	0.455 / 0.471
PatchTST	0.409 / 0.425
TimesNet	0.435 / 0.443
Autoformer	0.423 / 0.439

ETTh2 $\rightarrow$ ETTm1	
Model	MSE / MAE
RDTU	0.412 / 0.434
Time-LLM	0.414 / 0.438
LLMTime	1.933 / 0.984
GPT4TS	0.769 / 0.567
DLinear	0.649 / 0.537
PatchTST	0.568 / 0.492
TimesNet	0.769 / 0.567
Autoformer	0.755 / 0.591

4.3 Few-shot Forecasting

Setups. LLMs have demonstrated remarkable few-shot learning capabilities [13]. To evaluate the data efficiency of our framework in low-resource scenarios, we conduct few-shot learning experiments by restricting the training set to only 10% of the available data.

Results. As reported in Table S3, RDTU exhibits exceptional few-shot capabilities, securing the best performance across all metrics (14 out of 14). While data scarcity severely impacts deep learning baselines like PatchTST (ETTh1 MSE 0.633), RDTU maintains a robust MSE of **0.551** on the same task. Moreover, it consistently outperforms adapter-based LLMs such as Time-LLM (0.556) and GPT4TS (0.590). This validates that our direct unification strategy effectively leverages the pre-trained backbone to mitigate overfitting, proving superior efficiency over complex auxiliary adapters in data-limited scenarios.

4.4 Zero-shot Forecasting

Setups. To assess the generalization capability of our framework under distribution shifts, we conduct zero-shot learning experiments using the ETT dataset family.

Results. As reported in Table S4, RDTU demonstrates exceptional transferability, achieving the lowest horizon-averaged MSE in **7 of 8 transfer directions**. For $ETTh2 \rightarrow ETTm1$, RDTU obtains 0.480 MSE compared with 0.479 for Time-LLM, while its MAE is lower (0.471 versus 0.474). It highlights a fundamental advantage over supervised baselines, significantly outperforming Time-LLM in the $ETTh1 \rightarrow ETTm2$ transfer (MSE **0.344** vs.

0.353). Moreover, RDTU consistently surpasses competing LLM-based methods; in the challenging $ETTh2 \rightarrow ETTh1$ scenario, it maintains a robust MSE of **0.480**, far exceeding GPT4TS (0.757). These results support cross-dataset transfer within the evaluated ETT-family settings.

4.5 Showcases Analysis

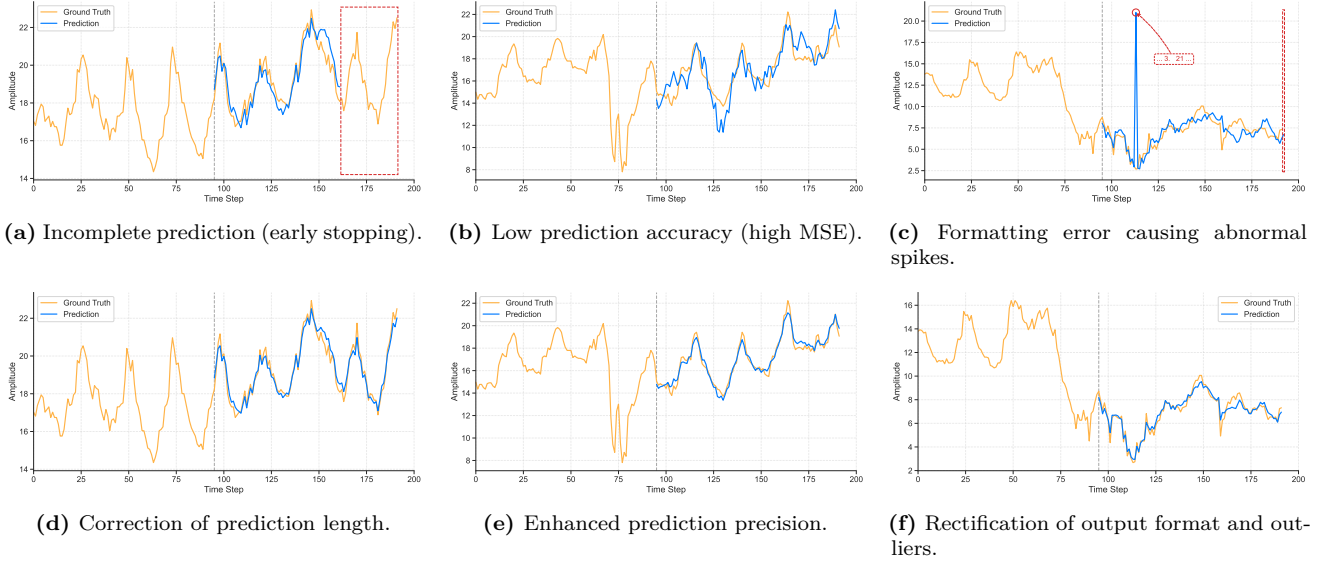


Figure S4: Showcase visualizations of input-96-predict-96 results on the ETTh1 dataset. The top row (a–c) displays the baseline performance of the STIT model, which exhibits limitations in sequence length, accuracy, and output formatting. The bottom row (d–f) shows the improvements achieved after applying the proposed RDRF framework. More showcases are provided in [Appendix E](#).

To intuitively understand how our proposed *Temporal Refinement Reward System* in RDRF stage refines the model’s behavior, we visualize a comparison between the STIT baseline and the final RDTU model in [Figure S4](#). This qualitative analysis targets three representative failure modes, validating the necessity of our composite reward system.

Correction of Prediction Length As observed in [Figure S4a](#), the STIT model frequently suffers from "early stopping," failing to track the generation count and terminating the sequence before reaching the required horizon. This indicates a lack of internal counting logic. In contrast, [Figure S4d](#) demonstrates the impact of the **Sequence Length Consistency Reward** (R_{len}). By imposing a soft exponential length constraint, the RDRF stage forces the model to attend to the output length, ensuring the generated sequence perfectly matches the target horizon without truncation.

Enhancement of Prediction Precision [Figure S4b](#) illustrates a case where the STIT model captures the general trend direction but fits loosely to the ground truth, exhibiting high variance and failing to capture local volatility. This corresponds to the limitations of token-level tuning in forecasting tasks. [Figure S4e](#) showcases the refinement achieved via the **Prediction Accuracy Reward** (R_{acc}). With its exponential scaling, R_{acc} penalizes even minor deviations in the high-precision regime, driving the model to produce a curve that hugs the ground truth tightly and significantly reducing the MSE.

Rectification of Formatting and Outliers Most critically, [Figure S4c](#) reveals a severe "Integer Projection" hallucination where the STIT model fails to generate the decimal point (e.g., splitting a value like ‘3.21’ into two tokens ‘3’ and ‘21’). This formatting error inserts an extreme outlier into the time series and causes a phase shift in all subsequent data points. [Figure S4f](#) confirms that the **Format Compliance Reward** (R_{fmt}) effectively eliminates this issue. By penalizing the model step-by-step for missing floating-point structures, the RDRF system enforces strict adherence to numerical syntax, restoring the smooth, continuous trajectory of the time series and preventing catastrophic outliers.

4.6 Ablation and Sensitivity Analysis

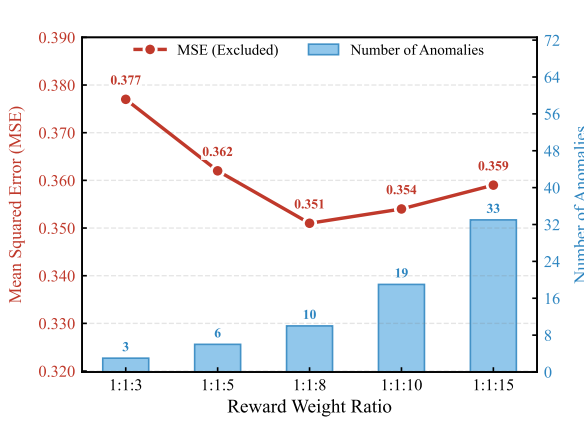
In this section, we perform comprehensive ablation studies to dissect the effectiveness of each component within the RDTU framework. Followed by [8], all experiments reported in this section are conducted on the **ETTh1** dataset with a fixed prediction horizon of $H = 96$ unless otherwise explicitly stated.

Component Analysis of the RDRF Framework. To validate our design, we conducted an ablation study

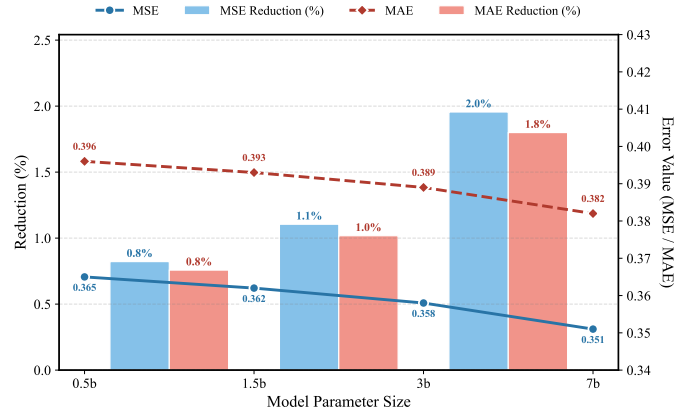
on the ETTh1 dataset. As shown in Table S5, while the off-the-shelf Qwen2.5-Instruct failed to generate valid outputs, the STIT model alone established a robust baseline with an MSE of **0.403**. This pivotal result demonstrates that complex pre-alignment modules are unnecessary; instruction tuning alone provides sufficient modality awareness for the LLM to function as a competent forecaster. Building on this foundation, the RDFR stage proved essential for precision. By imposing strict structural constraints and accuracy-driven rewards, the final RDTU framework further reduced the MSE to **0.351** with a **12.90%** improvement over the STIT baseline, confirming that reinforcement learning effectively bridges the gap between basic instruction following and high-precision regression.

Table S5: Ablation study of different reward components in the RDFR training stage. Improvements are computed relative to Qwen2.5-Instruct-STIT.

Method	MSE ↓	MAE ↓
Qwen2.5-Instruct (Zero-shot)	—	—
Qwen2.5-Instruct-STIT	0.403	0.428
+ R_{len}	0.401 (-0.50%)	0.425 (-0.70%)
+ R_{fmt}	0.395 (-1.99%)	0.419 (-2.10%)
+ R_{acc}	0.367 (-8.93%)	0.398 (-7.01%)
+ $R_{len} + R_{fmt}$	0.388 (-3.72%)	0.412 (-3.74%)
RDTU (R_{total})	0.351 (-12.90%)	0.382 (-10.75%)



(a) Impact of the reward weight ratio.



(b) Scaling laws on training efficiency and forecasting performance.

Figure S5: Ablation analysis of reward weighting and model scaling. Left: the red line represents the Mean Squared Error (MSE) on the excluded dataset, while the blue bars indicate the number of detected anomalies. Right: the line chart tracks MSE and MAE, showing that increased model capacity leads to consistent improvements in prediction accuracy.

Impact of Reward Weight Ratios. We investigate the optimal balance between structural constraints and regression precision by varying the accuracy reward weight λ_{acc} (fixing $\lambda_{len} : \lambda_{fmt}$ at 1 : 1). Results on ETTh1 identify 1 : 1 : 8 as the optimal configuration, achieving the lowest MSE of **0.351**. Deviating from this equilibrium degrades performance via two distinct failure modes: low weights (1 : 1 : 3) result in under-optimization of numerical values (MSE 0.377), while excessive weights (1 : 1 : 15) induce "structural collapse." As shown in Figure S5a, pushing beyond the optimal threshold causes formatting anomalies to spike from 10 to 33, demonstrating that overpowering syntax constraints with accuracy objectives ultimately corrupts prediction validity.

Impact of Model Scaling. We evaluate the impact of model capacity across the Qwen2.5 family (0.5B, 1.5B, 3B, 7B). Results in Figure S5b reveal a robust scaling law: as parameter count increases, forecasting accuracy consistently improves, with MSE on ETTh1 monotonically decreasing from 0.365 (0.5B) to 0.351 (7B). This confirms that **RDTU** effectively harnesses the stronger reasoning capabilities of larger backbones to achieve finer precision.

5 Conclusion and Limitations

In this paper, we proposed RDTU, a streamlined framework that directly formulates time series forecasting as a native token generation task without relying on complex pre-alignment modules. Through Supervised Temporal Instruction Tuning (STIT) for structure-aware forecasting adaptation and Reinforcement-Driven Forecasting Refinement (RDFR) for numerical precision and output validity, RDTU effectively transforms a general-purpose LLM into a reliable time series forecaster. Extensive experiments across long-term, short-term, few-shot, and zero-shot forecasting benchmarks demonstrate its strong accuracy and generalization ability. While RDTU achieves promising results on standard benchmarks, future work will further examine its robustness across broader real-world temporal domains.

A Experimental Details

A.1 Implementation

For **Stage I (STIT)**, we utilize LoRA ($r = 16, \alpha = 32$) targeting query and value modules. Optimization is performed via AdamW (learning rate 2×10^{-4} , global batch size 32) with a cosine annealing scheduler (0.03 warmup). In **Stage II (RDFR)**, we initialize the policy from the SFT checkpoint with a reduced learning rate of 5×10^{-6} and a KL-divergence coefficient $\beta_{kl} = 0.04$. We configure the Group Size $G = 8$, sampling 8 outputs per prompt. The total reward function R_{total} combines length consistency (R_{len}), format compliance (R_{fmt}), and prediction accuracy (R_{acc}) with a strict ratio of 1:1:8 ($\lambda_{len} = 0.1, \lambda_{fmt} = 0.1, \lambda_{acc} = 0.8$), prioritizing forecasting precision. Specifically, R_{fmt} aggregates sub-goals (number, float, precision) with weights $\{0.2, 0.3, 0.5\}$, while sensitivity factors are set to $\alpha = 0.5$ for length decay and $\beta = 10.0$ for accuracy scaling.

Training Budget and Resource Usage. To further clarify the practical cost of the proposed two-stage training pipeline, we report the per-dataset training budget and resource usage in Table S6. Although RDTU adopts a 7B LLM backbone, the overall training cost remains moderate. For the STIT stage, training on standard long-term forecasting benchmarks requires only about 2.5–4.3 hours, with a peak memory usage of approximately 24 GB per GPU. More importantly, the RDFR stage introduces only a small additional overhead, requiring about 0.3–0.5 hours and approximately 32 GB peak memory per GPU. This confirms that the proposed reinforcement-driven refinement is a lightweight yet effective stage: it improves numerical precision and output validity without introducing extra pre-alignment modules or substantial architectural overhead.

Table S6: Training settings and resource usage on different datasets.

Dataset	Stage	Training size	Epochs	Hours	VRAM / GPU
ETTh1	STIT / RDFR	8,545 / 425	15 / 3	~2.5 / ~0.3	~24 GB / ~32 GB
ETTh2	STIT / RDFR	8,545 / 425	15 / 3	~2.5 / ~0.3	~24 GB / ~32 GB
ETTM1	STIT / RDFR	34,465 / 1,713	5 / 2	~4.3 / ~0.5	~24 GB / ~32 GB
ETTM2	STIT / RDFR	34,465 / 1,713	5 / 2	~4.3 / ~0.5	~24 GB / ~32 GB
Electricity	STIT / RDFR	18,317 / 911	8 / 3	~3.7 / ~0.5	~24 GB / ~32 GB
Traffic	STIT / RDFR	12,185 / 606	10 / 3	~3.0 / ~0.3	~24 GB / ~32 GB
Weather	STIT / RDFR	36,792 / 1,830	4 / 2	~3.7 / ~0.5	~24 GB / ~32 GB

A.2 Evaluation Metrics

For evaluation metrics, we utilize the mean square error (MSE) and mean absolute error (MAE) for long-term forecasting. In terms of the short-term forecasting on M4 benchmark, we adopt the symmetric mean absolute percentage error (SMAPE), mean absolute scaled error (MASE), and overall weighted average (OWA) as in N-BEATS [20]. Note that OWA is a specific metric utilized in the M4 competition. The calculations of these

metrics are as follows:

$$\begin{aligned}
\text{MSE} &= \frac{1}{H} \sum_{h=1}^H (\mathbf{Y}_h - \hat{\mathbf{Y}}_h)^2, & \text{MAE} &= \frac{1}{H} \sum_{h=1}^H |\mathbf{Y}_h - \hat{\mathbf{Y}}_h|, \\
\text{SMAPE} &= \frac{200}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h| + |\hat{\mathbf{Y}}_h|}, & \text{MAPE} &= \frac{100}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h|}, \\
\text{MASE} &= \frac{1}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{\frac{1}{H-s} \sum_{j=s+1}^H |\mathbf{Y}_j - \mathbf{Y}_{j-s}|}, & \text{OWA} &= \frac{1}{2} \left[\frac{\text{SMAPE}}{\text{SMAPE}_{\text{Naive2}}} + \frac{\text{MASE}}{\text{MASE}_{\text{Naive2}}} \right],
\end{aligned}$$

where s is the periodicity of the time series data. H denotes the number of data points (i.e., prediction horizon in our cases). $\mathbf{Y}_h$ and $\hat{\mathbf{Y}}_h$ are the h -th ground truth and prediction where $h \in \{1, \dots, H\}$.

Declaration of LLM Usage. LLMs are a core methodological component of this work. Specifically, we use Qwen2.5-Instruct as the backbone forecasting model and adapt it to time series forecasting through the proposed two-stage RDTU framework, including Supervised Temporal Instruction Tuning (STIT) and Reinforcement-Driven Forecasting Refinement (RDFR). The LLM is used to generate numerical time series predictions under structured prompts, while the training procedure, reward design, hyperparameters, and evaluation settings are documented in the paper and appendix.

A.3 Dataset Details

Table S7: Dataset statistics are from [27]. The dimension indicates the number of time series (i.e., channels), and the dataset size is organized in (training, validation, testing).

Dataset	Dim.	Horizons	Train / val. / test	Frequency	Domain
ETTm1	7	[96, 192, 336, 720]	(34465, 11521, 11521)	15 min	Temperature
ETTm2	7	[96, 192, 336, 720]	(34465, 11521, 11521)	15 min	Temperature
ETTh1	7	[96, 192, 336, 720]	(8545, 2881, 2881)	1 hour	Temperature
ETTh2	7	[96, 192, 336, 720]	(8545, 2881, 2881)	1 hour	Temperature
Electricity	321	[96, 192, 336, 720]	(18317, 2633, 5261)	1 hour	Electricity
Traffic	862	[96, 192, 336, 720]	(12185, 1757, 3509)	1 hour	Transportation
Weather	21	[96, 192, 336, 720]	(36792, 5271, 10540)	10 min	Weather
ILI	7	[24, 36, 48, 60]	(617, 74, 170)	1 week	Illness
M4-Yearly	1	6	(23000, 0, 23000)	Yearly	Demographic
M4-Quarterly	1	8	(24000, 0, 24000)	Quarterly	Finance
M4-Monthly	1	18	(48000, 0, 48000)	Monthly	Industry
M4-Weekly	1	13	(359, 0, 359)	Weekly	Macro
M4-Daily	1	14	(4227, 0, 4227)	Daily	Micro
M4-Hourly	1	48	(414, 0, 414)	Hourly	Other

A.4 Statistical Significance Analysis

To examine whether the main performance gains are statistically reliable, we repeat RDTU and the strongest language-based baseline, Time-LLM, under three independent random seeds on representative long-term forecasting settings. We select ETTh1, Electricity, and Traffic because they cover different data scales and domains, including temperature, electricity consumption, and transportation. As shown in Table S8, RDTU consistently achieves lower MSE and MAE across all selected datasets. The paired two-sided t -test over matched dataset-seed pairs yields statistically significant improvements, with average p -values of 0.013 for MSE and 0.015 for MAE. These results indicate that the observed superiority of RDTU over language-based baselines is stable across random seeds rather than being caused by incidental initialization effects.

We further evaluate the statistical reliability of the RDFR stage, which is the key refinement component of RDTU. Specifically, we repeat both the STIT baseline and the final RDTU model under three independent random seeds on ETTh1 with $H = 96$. As shown in Table S9, RDFR consistently improves the STIT baseline across all runs, reducing MSE from 0.403 ± 0.002 to 0.351 ± 0.002 and MAE from 0.428 ± 0.002 to 0.382 ± 0.002 . The paired two-sided t -test gives $p = 0.004$ for MSE and $p = 0.006$ for MAE, confirming that the performance gain introduced

Table S8: Statistical significance analysis on representative long-term forecasting settings with $H = 96$. We report mean and standard deviation over three independent runs with different random seeds. p -values are computed using a paired two-sided t -test between RDTU and Time-LLM over matched dataset-seed pairs.

Dataset	Metric	RDTU	Time-LLM	Gain	p-value
ETTh1	MSE	0.352 ± 0.003	0.376 ± 0.004	6.38%	0.018
ETTh1	MAE	0.383 ± 0.002	0.403 ± 0.003	4.96%	0.021
Electricity	MSE	0.125 ± 0.002	0.137 ± 0.002	8.76%	0.014
Electricity	MAE	0.211 ± 0.002	0.233 ± 0.003	9.44%	0.012
Traffic	MSE	0.359 ± 0.004	0.393 ± 0.005	8.65%	0.011
Traffic	MAE	0.236 ± 0.003	0.268 ± 0.004	11.94%	0.009
Average	MSE	0.279 ± 0.003	0.302 ± 0.004	7.62%	0.013
Average	MAE	0.277 ± 0.002	0.301 ± 0.003	7.97%	0.015

Table S9: Statistical significance analysis of the RDFR stage on ETTh1 with $H = 96$. We report mean and standard deviation over three independent runs. p -values are computed using a paired two-sided t -test between STIT and RDTU.

Method	MSE ↓	MAE ↓	MSE Gain	MAE Gain
Qwen2.5-Instruct-STIT	0.403 ± 0.002	0.428 ± 0.002	–	–
RDTU w/ RDFR	0.351 ± 0.002	0.382 ± 0.002	12.90%	10.75%
p -value	0.004	0.006	–	–

by RDFR is statistically significant. This supports our claim that RDFR reliably bridges the gap between basic instruction-following ability and high-precision numerical forecasting.

B Long-term and Short-term Forecasting

B.1 Long-term Forecasting

By leveraging two-stage RDTU framework while bypassing pre-alignment, our method attains SOTA performance in **35** instances across eight time series benchmarks. This underscores the considerable potential of LLMs as robust and reliable time series forecasters.

Table S10 presents a comprehensive evaluation of long-term forecasting performance across eight benchmark datasets, categorizing baseline models by their input modality (Language, Multi-Modal, Visual, and Numerical).

Table S10: Full long-term forecasting results. Each entry is MSE / MAE; lower values are better.

ETTh1					
Model	96	192	336	720	Avg
RDTU	0.351 / 0.382	0.389 / 0.403	0.412 / 0.419	0.442 / 0.446	0.399 / 0.413
GPT4TS	0.370 / 0.389	0.412 / 0.413	0.448 / 0.431	0.441 / 0.449	0.418 / 0.421
Time-LLM	0.376 / 0.402	0.407 / 0.421	0.430 / 0.438	0.457 / 0.468	0.418 / 0.432
DMMV-A	0.354 / 0.389	0.393 / 0.405	0.387 / 0.413	0.445 / 0.450	0.395 / 0.414
Time-VLM	0.361 / 0.386	0.397 / 0.415	0.420 / 0.421	0.441 / 0.458	0.405 / 0.420
VisionTS	0.355 / 0.386	0.395 / 0.407	0.419 / 0.421	0.458 / 0.460	0.407 / 0.419
PatchTST	0.370 / 0.399	0.413 / 0.421	0.422 / 0.436	0.447 / 0.466	0.413 / 0.431
CycleNet	0.374 / 0.396	0.406 / 0.415	0.431 / 0.430	0.450 / 0.464	0.415 / 0.426
TimesNet	0.384 / 0.402	0.436 / 0.429	0.491 / 0.469	0.521 / 0.500	0.458 / 0.450
DLinear	0.375 / 0.399	0.405 / 0.416	0.439 / 0.416	0.472 / 0.490	0.423 / 0.430
FEDformer	0.376 / 0.419	0.420 / 0.448	0.459 / 0.465	0.506 / 0.507	0.440 / 0.460

Table S10 (continued). ETTh2

ETTh2					
Model	96	192	336	720	Avg
RDTU	0.263 / 0.329	0.334 / 0.372	0.355 / 0.381	0.395 / 0.425	0.337 / 0.377
GPT4TS	0.280 / 0.335	0.348 / 0.380	0.380 / 0.405	0.406 / 0.436	0.354 / 0.389
Time-LLM	0.286 / 0.346	0.361 / 0.391	0.390 / 0.414	0.405 / 0.434	0.361 / 0.396
DMMV-A	0.294 / 0.349	0.339 / 0.395	0.322 / 0.384	0.392 / 0.425	0.337 / 0.388
Time-VLM	0.267 / 0.335	0.326 / 0.373	0.357 / 0.406	0.412 / 0.449	0.341 / 0.391
VisionTS	0.288 / 0.334	0.349 / 0.380	0.364 / 0.398	0.403 / 0.431	0.351 / 0.386
PatchTST	0.274 / 0.336	0.339 / 0.379	0.329 / 0.380	0.379 / 0.422	0.330 / 0.379
CycleNet	0.279 / 0.341	0.342 / 0.385	0.371 / 0.413	0.426 / 0.451	0.355 / 0.398
TimesNet	0.340 / 0.374	0.402 / 0.414	0.452 / 0.452	0.462 / 0.468	0.414 / 0.427
DLinear	0.289 / 0.353	0.383 / 0.418	0.448 / 0.465	0.605 / 0.551	0.431 / 0.447
FEDformer	0.358 / 0.397	0.429 / 0.439	0.496 / 0.487	0.463 / 0.474	0.437 / 0.449

Table S10 (continued). ETTm1

ETTh1					
Model	96	192	336	720	Avg
RDTU	0.272 / 0.326	0.327 / 0.354	0.350 / 0.375	0.410 / 0.412	0.340 / 0.367
GPT4TS	0.300 / 0.340	0.343 / 0.368	0.376 / 0.386	0.431 / 0.416	0.363 / 0.378
Time-LLM	0.291 / 0.341	0.341 / 0.369	0.359 / 0.379	0.433 / 0.419	0.356 / 0.377
DMMV-A	0.279 / 0.329	0.317 / 0.357	0.351 / 0.381	0.411 / 0.415	0.340 / 0.371
Time-VLM	0.304 / 0.346	0.332 / 0.366	0.364 / 0.383	0.402 / 0.410	0.351 / 0.376
VisionTS	0.284 / 0.332	0.327 / 0.362	0.354 / 0.382	0.411 / 0.415	0.344 / 0.373
PatchTST	0.290 / 0.342	0.332 / 0.369	0.366 / 0.392	0.416 / 0.420	0.351 / 0.381
CycleNet	0.299 / 0.348	0.334 / 0.367	0.368 / 0.386	0.417 / 0.414	0.355 / 0.379
TimesNet	0.338 / 0.375	0.374 / 0.387	0.410 / 0.411	0.478 / 0.450	0.400 / 0.406
DLinear	0.299 / 0.343	0.335 / 0.365	0.369 / 0.386	0.425 / 0.421	0.357 / 0.379
FEDformer	0.379 / 0.419	0.426 / 0.441	0.445 / 0.459	0.543 / 0.490	0.448 / 0.452

Table S10 (continued). ETTm2

ETTm2					
Model	96	192	336	720	Avg
RDTU	0.157 / 0.244	0.219 / 0.288	0.270 / 0.323	0.354 / 0.379	0.250 / 0.309
GPT4TS	0.163 / 0.249	0.222 / 0.291	0.273 / 0.327	0.357 / 0.376	0.254 / 0.311
Time-LLM	0.162 / 0.248	0.235 / 0.304	0.280 / 0.329	0.366 / 0.382	0.261 / 0.316
DMMV-A	0.172 / 0.260	0.227 / 0.298	0.272 / 0.327	0.351 / 0.381	0.256 / 0.317
Time-VLM	0.160 / 0.250	0.215 / 0.291	0.270 / 0.325	0.348 / 0.378	0.248 / 0.311
VisionTS	0.174 / 0.262	0.228 / 0.297	0.281 / 0.337	0.384 / 0.410	0.267 / 0.327
PatchTST	0.165 / 0.255	0.220 / 0.292	0.274 / 0.329	0.362 / 0.385	0.255 / 0.315
CycleNet	0.159 / 0.247	0.214 / 0.286	0.269 / 0.322	0.363 / 0.382	0.251 / 0.309
TimesNet	0.187 / 0.267	0.249 / 0.309	0.321 / 0.351	0.408 / 0.403	0.291 / 0.333
DLinear	0.167 / 0.260	0.224 / 0.303	0.281 / 0.342	0.397 / 0.421	0.267 / 0.332
FEDformer	0.203 / 0.287	0.269 / 0.328	0.325 / 0.366	0.421 / 0.415	0.305 / 0.349

Table S10 (continued). Illness

Illness					
Model	24	36	48	60	Avg
RDTU	1.609 / 0.771	1.453 / 0.835	1.545 / 0.841	1.630 / 0.802	1.559 / 0.812
GPT4TS	1.869 / 0.823	1.853 / 0.854	1.886 / 0.855	1.877 / 0.877	1.871 / 0.852
Time-LLM	1.792 / 0.807	1.833 / 0.833	2.269 / 1.012	2.177 / 0.925	2.018 / 0.894
DMMV-A	1.409 / 0.754	1.290 / 0.745	1.499 / 0.810	1.428 / 0.773	1.407 / 0.771
Time-VLM	—	—	—	—	—
VisionTS	1.613 / 0.834	1.316 / 0.750	1.548 / 0.818	1.450 / 0.783	1.482 / 0.796
PatchTST	1.319 / 0.754	1.430 / 0.834	1.553 / 0.815	1.470 / 0.788	1.443 / 0.798
CycleNet	2.255 / 1.017	2.121 / 0.950	2.187 / 1.007	2.185 / 0.997	2.187 / 0.992
TimesNet	2.317 / 0.934	1.972 / 0.920	2.238 / 0.940	2.027 / 0.928	2.139 / 0.931
DLinear	2.215 / 1.081	1.963 / 0.963	2.130 / 1.024	2.368 / 1.096	2.169 / 1.041
FEDformer	3.228 / 1.260	2.679 / 1.080	2.622 / 1.078	2.857 / 1.157	2.847 / 1.144

Table S10 (continued). Electricity

Electricity					
Model	96	192	336	720	Avg
RDTU	0.124 / 0.210	0.144 / 0.235	0.162 / 0.257	0.193 / 0.281	0.156 / 0.246
GPT4TS	0.141 / 0.239	0.158 / 0.253	0.172 / 0.266	0.207 / 0.293	0.170 / 0.263
Time-LLM	0.137 / 0.233	0.152 / 0.247	0.169 / 0.267	0.200 / 0.290	0.165 / 0.259
DMMV-A	0.126 / 0.213	0.145 / 0.237	0.162 / 0.254	0.197 / 0.286	0.158 / 0.248
Time-VLM	0.142 / 0.245	0.157 / 0.260	0.174 / 0.276	0.214 / 0.308	0.172 / 0.272
VisionTS	0.127 / 0.217	0.148 / 0.237	0.163 / 0.253	0.199 / 0.293	0.159 / 0.250
PatchTST	0.129 / 0.222	0.157 / 0.240	0.163 / 0.259	0.197 / 0.290	0.162 / 0.253
CycleNet	0.128 / 0.223	0.144 / 0.237	0.160 / 0.254	0.198 / 0.287	0.158 / 0.250
TimesNet	0.168 / 0.272	0.184 / 0.289	0.198 / 0.300	0.220 / 0.320	0.193 / 0.295
DLinear	0.140 / 0.237	0.153 / 0.249	0.169 / 0.267	0.203 / 0.301	0.166 / 0.264
FEDformer	0.193 / 0.308	0.201 / 0.315	0.214 / 0.329	0.246 / 0.355	0.214 / 0.327

Table S10 (continued). Weather

Weather					
Model	96	192	336	720	Avg
RDTU	0.141 / 0.184	0.189 / 0.227	0.239 / 0.272	0.314 / 0.322	0.221 / 0.251
GPT4TS	0.148 / 0.188	0.192 / 0.230	0.246 / 0.273	0.320 / 0.328	0.227 / 0.255
Time-LLM	0.155 / 0.199	0.223 / 0.261	0.251 / 0.279	0.345 / 0.342	0.244 / 0.270
DMMV-A	0.143 / 0.195	0.187 / 0.242	0.237 / 0.273	0.302 / 0.315	0.217 / 0.256
Time-VLM	0.148 / 0.200	0.193 / 0.240	0.243 / 0.281	0.312 / 0.332	0.224 / 0.263
VisionTS	0.146 / 0.191	0.194 / 0.238	0.243 / 0.275	0.318 / 0.328	0.225 / 0.258
PatchTST	0.149 / 0.198	0.194 / 0.241	0.245 / 0.282	0.314 / 0.334	0.226 / 0.264
CycleNet	0.167 / 0.221	0.212 / 0.258	0.260 / 0.293	0.328 / 0.339	0.242 / 0.278
TimesNet	0.172 / 0.220	0.219 / 0.261	0.280 / 0.306	0.365 / 0.359	0.259 / 0.287
DLinear	0.176 / 0.237	0.220 / 0.282	0.265 / 0.319	0.333 / 0.362	0.249 / 0.300
FEDformer	0.217 / 0.296	0.276 / 0.336	0.339 / 0.380	0.403 / 0.428	0.309 / 0.360

Table S10 (continued). Traffic

Traffic					
Model	96	192	336	720	Avg
RDTU	0.358 / 0.235	0.375 / 0.248	0.388 / 0.256	0.435 / 0.280	0.389 / 0.255
GPT4TS	0.396 / 0.264	0.412 / 0.268	0.421 / 0.273	0.455 / 0.291	0.421 / 0.274
Time-LLM	0.392 / 0.267	0.409 / 0.271	0.434 / 0.296	0.451 / 0.291	0.422 / 0.281
DMMV-A	0.344 / 0.237	0.363 / 0.249	0.387 / 0.256	0.433 / 0.284	0.389 / 0.257
Time-VLM	0.393 / 0.290	0.405 / 0.296	0.420 / 0.305	0.459 / 0.323	0.419 / 0.304
VisionTS	0.346 / 0.232	0.376 / 0.245	0.389 / 0.252	0.432 / 0.293	0.386 / 0.256
PatchTST	0.360 / 0.249	0.379 / 0.256	0.392 / 0.264	0.432 / 0.286	0.391 / 0.264
CycleNet	0.397 / 0.278	0.411 / 0.283	0.424 / 0.289	0.450 / 0.305	0.421 / 0.289
TimesNet	0.593 / 0.321	0.617 / 0.336	0.629 / 0.336	0.640 / 0.350	0.620 / 0.336
DLinear	0.410 / 0.282	0.423 / 0.287	0.436 / 0.296	0.466 / 0.315	0.434 / 0.295
FEDformer	0.587 / 0.366	0.604 / 0.373	0.621 / 0.383	0.626 / 0.382	0.610 / 0.376

Source-reported ranking counts. RDTU: 35, GPT4TS: 2, Time-LLM: 0, DMMV-A: 23, Time-VLM: 6, VisionTS: 6, PatchTST: 7, CycleNet: 6, TimesNet: 0, DLinear: 0, FEDformer: 0. The original counting convention is not specified; these values are retained without recalculation.

Table S11: Long-term forecasting: additional baselines. Each entry is MSE / MAE; lower values are better.

ETTh1					
Model	96	192	336	720	Avg
RDTU	0.351 / 0.382	0.389 / 0.403	0.412 / 0.419	0.442 / 0.446	0.399 / 0.413
Autoformer	0.449 / 0.459	0.500 / 0.482	0.521 / 0.496	0.514 / 0.512	0.496 / 0.487
Stationary	0.513 / 0.491	0.534 / 0.504	0.588 / 0.535	0.643 / 0.616	0.570 / 0.537
ETSformer	0.494 / 0.479	0.538 / 0.504	0.574 / 0.521	0.562 / 0.535	0.542 / 0.510
LightTS	0.424 / 0.432	0.475 / 0.462	0.518 / 0.488	0.547 / 0.533	0.491 / 0.479
Informer	0.865 / 0.713	1.008 / 0.792	1.107 / 0.809	1.181 / 0.865	1.040 / 0.795
Reformer	0.837 / 0.728	0.923 / 0.766	1.097 / 0.835	1.257 / 0.889	1.029 / 0.805

Table S11 (continued). ETTh2

ETTh2					
Model	96	192	336	720	Avg
RDTU	0.263 / 0.329	0.334 / 0.372	0.355 / 0.381	0.395 / 0.425	0.337 / 0.377
Autoformer	0.346 / 0.388	0.456 / 0.452	0.482 / 0.486	0.515 / 0.511	0.450 / 0.459
Stationary	0.476 / 0.458	0.512 / 0.493	0.552 / 0.551	0.562 / 0.560	0.526 / 0.516
ETSformer	0.340 / 0.391	0.430 / 0.439	0.485 / 0.479	0.500 / 0.497	0.439 / 0.452
LightTS	0.397 / 0.437	0.520 / 0.504	0.626 / 0.559	0.863 / 0.672	0.602 / 0.543
Informer	3.755 / 1.525	5.602 / 1.931	4.721 / 1.835	3.647 / 1.625	4.431 / 1.729
Reformer	2.626 / 1.317	11.12 / 2.979	9.323 / 2.769	3.874 / 1.697	6.736 / 2.191

Table S11 (continued). ETTm1

ETTm1					
Model	96	192	336	720	Avg
RDTU	0.272 / 0.326	0.327 / 0.354	0.350 / 0.375	0.410 / 0.412	0.340 / 0.367
Autoformer	0.505 / 0.475	0.553 / 0.496	0.621 / 0.537	0.671 / 0.561	0.588 / 0.517
Stationary	0.386 / 0.398	0.459 / 0.444	0.495 / 0.464	0.585 / 0.516	0.481 / 0.456
ETSformer	0.375 / 0.398	0.408 / 0.410	0.435 / 0.428	0.499 / 0.462	0.429 / 0.425
LightTS	0.374 / 0.400	0.400 / 0.407	0.438 / 0.438	0.527 / 0.502	0.435 / 0.437
Informer	0.672 / 0.571	0.795 / 0.669	1.212 / 0.871	1.166 / 0.823	0.961 / 0.734
Reformer	0.538 / 0.528	0.658 / 0.592	0.898 / 0.721	1.102 / 0.841	0.799 / 0.671

Table S11 (continued). ETTm2

ETTm2					
Model	96	192	336	720	Avg
RDTU	0.157 / 0.244	0.219 / 0.288	0.270 / 0.323	0.354 / 0.379	0.250 / 0.309
Autoformer	0.255 / 0.339	0.281 / 0.340	0.339 / 0.372	0.433 / 0.432	0.327 / 0.371
Stationary	0.192 / 0.274	0.280 / 0.339	0.334 / 0.361	0.417 / 0.413	0.306 / 0.347
ETSformer	0.189 / 0.280	0.253 / 0.319	0.314 / 0.357	0.414 / 0.413	0.293 / 0.342
LightTS	0.209 / 0.308	0.311 / 0.382	0.442 / 0.466	0.675 / 0.587	0.409 / 0.436
Informer	0.365 / 0.453	0.533 / 0.563	1.363 / 0.887	3.379 / 1.338	1.410 / 0.810
Reformer	0.658 / 0.619	1.078 / 0.827	1.549 / 0.972	2.631 / 1.242	1.479 / 0.915

Table S11 (continued). Weather

Weather					
Model	96	192	336	720	Avg
RDTU	0.141 / 0.184	0.189 / 0.227	0.239 / 0.272	0.314 / 0.322	0.221 / 0.251
Autoformer	0.266 / 0.336	0.307 / 0.367	0.359 / 0.395	0.419 / 0.428	0.338 / 0.382
Stationary	0.173 / 0.223	0.245 / 0.285	0.321 / 0.338	0.414 / 0.410	0.288 / 0.314
ETSformer	0.197 / 0.281	0.237 / 0.312	0.298 / 0.353	0.352 / 0.288	0.271 / 0.334
LightTS	0.182 / 0.242	0.227 / 0.287	0.282 / 0.334	0.352 / 0.386	0.261 / 0.312
Informer	0.300 / 0.384	0.598 / 0.544	0.578 / 0.523	1.059 / 0.741	0.634 / 0.548
Reformer	0.689 / 0.596	0.752 / 0.638	0.639 / 0.596	1.130 / 0.792	0.803 / 0.656

Table S11 (continued). Electricity

Electricity					
Model	96	192	336	720	Avg
RDTU	0.124 / 0.210	0.144 / 0.235	0.162 / 0.257	0.193 / 0.281	0.156 / 0.246
Autoformer	0.201 / 0.317	0.222 / 0.334	0.231 / 0.338	0.254 / 0.361	0.227 / 0.338
Stationary	0.169 / 0.273	0.182 / 0.286	0.200 / 0.304	0.222 / 0.321	0.193 / 0.296
ETSformer	0.187 / 0.304	0.199 / 0.315	0.212 / 0.329	0.233 / 0.345	0.208 / 0.323
LightTS	0.207 / 0.307	0.213 / 0.316	0.230 / 0.333	0.265 / 0.360	0.229 / 0.329
Informer	0.274 / 0.368	0.296 / 0.386	0.300 / 0.394	0.373 / 0.439	0.311 / 0.397
Reformer	0.312 / 0.402	0.348 / 0.433	0.350 / 0.433	0.340 / 0.420	0.338 / 0.422

Table S11 (continued). Traffic

Traffic					
Model	96	192	336	720	Avg
RDTU	0.358 / 0.235	0.375 / 0.248	0.388 / 0.256	0.435 / 0.280	0.389 / 0.255
Autoformer	0.613 / 0.388	0.616 / 0.382	0.622 / 0.337	0.660 / 0.408	0.628 / 0.379
Stationary	0.612 / 0.338	0.613 / 0.340	0.618 / 0.328	0.653 / 0.355	0.624 / 0.340
ETSformer	0.607 / 0.392	0.621 / 0.399	0.622 / 0.396	0.632 / 0.396	0.621 / 0.396
LightTS	0.615 / 0.391	0.601 / 0.382	0.613 / 0.386	0.658 / 0.407	0.622 / 0.392
Informer	0.719 / 0.391	0.696 / 0.379	0.777 / 0.420	0.864 / 0.472	0.764 / 0.416
Reformer	0.732 / 0.423	0.733 / 0.420	0.742 / 0.420	0.755 / 0.423	0.741 / 0.422

Table S11 (continued). ILI

ILI					
Model	24	36	48	60	Avg
RDTU	1.609 / 0.771	1.453 / 0.835	1.545 / 0.841	1.630 / 0.802	1.559 / 0.812
Autoformer	3.483 / 1.287	3.103 / 1.148	2.669 / 1.085	2.770 / 1.125	3.006 / 1.161
Stationary	2.294 / 0.945	1.825 / 0.848	2.010 / 0.900	2.178 / 0.963	2.077 / 0.914
ETSformer	2.527 / 1.020	2.615 / 1.007	2.359 / 0.972	2.487 / 1.016	2.497 / 1.004
LightTS	8.313 / 2.144	6.631 / 1.902	7.299 / 1.982	7.283 / 1.985	7.382 / 2.003
Informer	5.764 / 1.677	4.755 / 1.467	4.763 / 1.469	5.264 / 1.564	5.137 / 1.544
Reformer	4.400 / 1.382	4.783 / 1.448	4.832 / 1.465	4.882 / 1.483	4.724 / 1.445

Source-reported ranking counts. RDTU: 40, Autoformer: 0, Stationary: 0, ETSformer: 0, LightTS: 0, Informer: 0, Reformer: 0. The original counting convention is not specified; these values are retained without recalculation.

Table S11 presents an extensive evaluation of the proposed RDTU method against six state-of-the-art baseline models (Autoformer, Stationary, ETSformer, LightTS, Informer, and Reformer) across eight benchmark datasets. The experimental results clearly demonstrate the superior performance of RDTU for long-term time series forecasting.

As indicated by the bold red values, RDTU achieves the lowest Mean Squared Error (MSE) across the listed settings and the lowest Mean Absolute Error (MAE) in all but one individual horizon setting ($H \in \{24, 36, 48, 60\}$ for ILI and $H \in \{96, 192, 336, 720\}$ for other datasets). The quantitative summary at the bottom of the table confirms this dominance, with a source-reported "1st Count" of 40 for RDTU and 0 for each listed baseline. This count is retained as reported; it is distinct from the number of individual MSE and MAE cells in the table. While models such as LightTS and ETSformer occasionally yield the second-best results (highlighted in blue), RDTU consistently provides significant accuracy improvements.

B.2 Short-term Forecasting

Our complete results on short-term forecasting are presented in Table S12.

Table S12 extends the analysis to short-term time series forecasting. Evaluation metrics including SMAPE, MASE, and OWA indicate that RDTU maintains its leading position. In the weighted average analysis across all sampling intervals (Yearly, Quarterly, Monthly, Others), RDTU records the best performance (e.g., Average OWA of 0.859), and matches Time-LLM on OWA while achieving lower SMAPE (OWA 0.859), significantly surpassing traditional methods like N-HiTS and N-BEATS.

Table S12: Full M4 short-term forecasting results. Lower values are better.

Yearly			
Model	SMAPE	MASE	OWA
RDTU	13.408	3.014	0.788
Time-LLM	13.419	3.005	0.789
GPT4TS	15.11	3.565	0.911
TimesNet	15.378	3.554	0.918
PatchTST	13.477	3.019	0.792
N-HiTS	13.422	3.056	0.795
N-BEATS	13.487	3.036	0.795
ETSformer	18.009	4.487	1.115
LightTS	14.247	3.109	0.827
DLinear	16.965	4.283	1.058
FEDformer	14.021	3.036	0.811
Stationary	13.717	3.078	0.807
Autoformer	13.974	3.134	0.822
Informer	14.727	3.418	0.881
Reformer	16.169	3.800	0.973

Table S12 (continued). Quarterly

Quarterly			
Model	SMAPE	MASE	OWA
RDTU	10.094	1.176	0.888
Time-LLM	10.110	1.178	0.889
GPT4TS	10.597	1.253	0.938
TimesNet	10.465	1.227	0.923
PatchTST	10.38	1.233	0.921
N-HiTS	10.185	1.18	0.893
N-BEATS	10.564	1.252	0.936
ETSformer	13.376	1.906	1.302
LightTS	11.364	1.328	1.000
DLinear	12.145	1.520	1.106
FEDformer	11.1	1.35	0.996
Stationary	10.958	1.325	0.981
Autoformer	11.338	1.365	1.012
Informer	11.360	1.401	1.027
Reformer	13.313	1.775	1.252

Table S12 (continued). Monthly

Monthly			
Model	SMAPE	MASE	OWA
RDTU	12.977	0.968	0.903
Time-LLM	12.980	0.963	0.903
GPT4TS	13.258	1.003	0.931
TimesNet	13.513	1.039	0.957
PatchTST	12.959	0.97	0.905
N-HiTS	13.059	1.013	0.929
N-BEATS	13.089	0.996	0.922
ETSformer	14.588	1.368	1.149
LightTS	14.014	1.053	0.981
DLinear	13.514	1.037	0.956
FEDformer	14.403	1.147	1.038
Stationary	13.917	1.097	0.998
Autoformer	13.958	1.103	1.002
Informer	14.062	1.141	1.024
Reformer	20.128	2.614	1.927

Table S12 (continued). Others

Others			
Model	SMAPE	MASE	OWA
RDTU	4.730	3.086	0.985
Time-LLM	4.795	3.178	1.006
GPT4TS	6.124	4.116	1.259
TimesNet	6.913	4.507	1.438
PatchTST	4.952	3.347	1.049
N-HiTS	4.711	3.054	0.977
N-BEATS	6.599	4.43	1.393
ETSformer	7.267	5.240	1.591
LightTS	15.880	11.434	3.474
DLinear	6.709	4.953	1.487
FEDformer	7.148	4.041	1.389
Stationary	6.302	4.064	1.304
Autoformer	5.485	3.865	1.187
Informer	24.460	20.960	5.879
Reformer	32.491	33.355	8.679

Table S12 (continued). Average

Average			
Model	SMAPE	MASE	OWA
RDTU	11.979	1.599	0.859
Time-LLM	11.983	1.595	0.859
GPT4TS	12.69	1.808	0.94
TimesNet	12.88	1.836	0.955
PatchTST	12.059	1.623	0.869
N-HiTS	12.035	1.625	0.869
N-BEATS	12.25	1.698	0.896
ETSformer	14.718	2.408	1.172
LightTS	13.525	2.111	1.051
DLinear	13.639	2.095	1.051
FEDformer	13.16	1.775	0.949
Stationary	12.780	1.756	0.930
Autoformer	12.909	1.771	0.939
Informer	14.086	2.718	1.230
Reformer	18.200	4.223	1.775

C Few-Shot and Zero-Shot Forecasting

C.1 Few-Shot Forecasting

Table S13: Full few-shot forecasting results: 10% training data. Each entry is MSE / MAE; lower values are better.

ETTh1					
Model	96	192	336	720	Avg
RDTU	0.443 / 0.451	0.482 / 0.479	0.585 / 0.536	0.694 / 0.587	0.551 / 0.513
Time-LLM	0.448 / 0.460	0.484 / 0.483	0.589 / 0.540	0.700 / 0.604	0.556 / 0.522
GPT4TS	0.458 / 0.456	0.570 / 0.516	0.608 / 0.535	0.725 / 0.591	0.590 / 0.525
DLinear	0.492 / 0.495	0.565 / 0.538	0.721 / 0.622	0.986 / 0.743	0.691 / 0.600
PatchTST	0.516 / 0.485	0.598 / 0.524	0.657 / 0.550	0.762 / 0.610	0.633 / 0.542
TimesNet	0.861 / 0.628	0.797 / 0.593	0.941 / 0.648	0.877 / 0.641	0.869 / 0.628
FEDformer	0.512 / 0.499	0.624 / 0.555	0.691 / 0.574	0.728 / 0.614	0.639 / 0.561
Autoformer	0.613 / 0.552	0.722 / 0.598	0.750 / 0.619	0.721 / 0.616	0.702 / 0.596
Stationary	0.918 / 0.639	0.915 / 0.629	0.939 / 0.644	0.887 / 0.645	0.915 / 0.639
ETSformer	1.112 / 0.806	1.155 / 0.823	1.179 / 0.832	1.273 / 0.874	1.180 / 0.834
LightTS	1.298 / 0.838	1.322 / 0.854	1.347 / 0.870	1.534 / 0.947	1.375 / 0.877
Informer	1.179 / 0.792	1.199 / 0.806	1.202 / 0.811	1.217 / 0.825	1.199 / 0.809
Reformer	1.184 / 0.790	1.295 / 0.850	1.294 / 0.854	1.223 / 0.838	1.249 / 0.833

Table S13 (continued). ETTh2

ETTh2					
Model	96	192	336	720	Avg
RDTU	0.271 / 0.322	0.369 / 0.367	0.402 / 0.423	0.425 / 0.443	0.367 / 0.389
Time-LLM	0.275 / 0.326	0.374 / 0.373	0.406 / 0.429	0.427 / 0.449	0.370 / 0.394
GPT4TS	0.331 / 0.374	0.402 / 0.411	0.406 / 0.433	0.449 / 0.464	0.397 / 0.421
DLinear	0.357 / 0.411	0.569 / 0.519	0.671 / 0.572	0.824 / 0.648	0.605 / 0.538
PatchTST	0.353 / 0.389	0.403 / 0.414	0.426 / 0.441	0.477 / 0.480	0.415 / 0.431
TimesNet	0.378 / 0.409	0.490 / 0.467	0.537 / 0.494	0.510 / 0.491	0.479 / 0.465
FEDformer	0.382 / 0.416	0.478 / 0.474	0.504 / 0.501	0.499 / 0.509	0.466 / 0.475
Autoformer	0.413 / 0.451	0.474 / 0.477	0.547 / 0.543	0.516 / 0.523	0.488 / 0.499
Stationary	0.389 / 0.411	0.473 / 0.455	0.507 / 0.480	0.477 / 0.472	0.462 / 0.455
ETSformer	0.678 / 0.619	0.785 / 0.666	0.839 / 0.694	1.273 / 0.874	0.894 / 0.713
LightTS	2.022 / 1.006	2.329 / 1.104	2.453 / 1.122	3.816 / 1.407	2.655 / 1.160
Informer	3.837 / 1.508	3.856 / 1.513	3.952 / 1.526	3.842 / 1.503	3.872 / 1.513
Reformer	3.788 / 1.533	3.552 / 1.483	3.395 / 1.526	3.205 / 1.401	3.485 / 1.486

Table S13 (continued). ETTm1

ETTm1					
Model	96	192	336	720	Avg
RDTU	0.343 / 0.386	0.375 / 0.411	0.410 / 0.424	0.478 / 0.473	0.402 / 0.424
Time-LLM	0.346 / 0.388	0.373 / 0.416	0.413 / 0.426	0.485 / 0.476	0.404 / 0.427
GPT4TS	0.390 / 0.404	0.429 / 0.423	0.469 / 0.439	0.569 / 0.498	0.464 / 0.441
DLinear	0.352 / 0.392	0.382 / 0.412	0.419 / 0.434	0.490 / 0.477	0.411 / 0.429
PatchTST	0.410 / 0.419	0.437 / 0.434	0.476 / 0.454	0.681 / 0.556	0.501 / 0.466
TimesNet	0.583 / 0.501	0.630 / 0.528	0.725 / 0.568	0.769 / 0.549	0.677 / 0.537
FEDformer	0.578 / 0.518	0.617 / 0.546	0.998 / 0.775	0.693 / 0.579	0.722 / 0.605
Autoformer	0.774 / 0.614	0.754 / 0.592	0.869 / 0.677	0.810 / 0.630	0.802 / 0.628
Stationary	0.761 / 0.568	0.781 / 0.574	0.803 / 0.587	0.844 / 0.581	0.797 / 0.578
ETSformer	0.911 / 0.688	0.955 / 0.703	0.991 / 0.719	1.062 / 0.747	0.980 / 0.714
LightTS	0.921 / 0.682	0.957 / 0.701	0.998 / 0.716	1.007 / 0.719	0.971 / 0.705
Informer	1.162 / 0.785	1.172 / 0.793	1.227 / 0.908	1.207 / 0.797	1.192 / 0.821
Reformer	1.442 / 0.847	1.444 / 0.862	1.450 / 0.866	1.366 / 0.850	1.426 / 0.856

Table S13 (continued). ETTm2

ETTm2					
Model	96	192	336	720	Avg
RDTU	0.174 / 0.260	0.236 / 0.308	0.272 / 0.323	0.413 / 0.384	0.274 / 0.319
Time-LLM	0.177 / 0.261	0.241 / 0.314	0.274 / 0.327	0.417 / 0.390	0.277 / 0.323
GPT4TS	0.188 / 0.269	0.251 / 0.309	0.307 / 0.346	0.426 / 0.417	0.293 / 0.335
DLinear	0.213 / 0.303	0.278 / 0.345	0.338 / 0.385	0.436 / 0.440	0.316 / 0.368
PatchTST	0.191 / 0.274	0.252 / 0.317	0.306 / 0.353	0.433 / 0.427	0.296 / 0.343
TimesNet	0.212 / 0.285	0.270 / 0.323	0.323 / 0.353	0.474 / 0.449	0.320 / 0.353
FEDformer	0.291 / 0.399	0.307 / 0.379	0.543 / 0.559	0.712 / 0.614	0.463 / 0.488
Autoformer	0.352 / 0.454	0.694 / 0.691	2.408 / 1.407	1.913 / 1.166	1.342 / 0.930
Stationary	0.229 / 0.308	0.291 / 0.343	0.348 / 0.376	0.461 / 0.438	0.332 / 0.366
ETSformer	0.331 / 0.430	0.400 / 0.464	0.469 / 0.498	0.589 / 0.557	0.447 / 0.487
LightTS	0.813 / 0.688	1.008 / 0.768	1.031 / 0.775	1.096 / 0.791	0.987 / 0.756
Informer	3.203 / 1.407	3.112 / 1.387	3.255 / 1.421	3.909 / 1.543	3.370 / 1.440
Reformer	4.195 / 1.628	4.042 / 1.601	3.963 / 1.585	3.711 / 1.532	3.978 / 1.587

Table S13 (continued). Weather

Weather					
Model	96	192	336	720	Avg
RDTU	0.159 / 0.209	0.199 / 0.244	0.257 / 0.292	0.311 / 0.335	0.232 / 0.270
Time-LLM	0.161 / 0.210	0.204 / 0.248	0.261 / 0.302	0.309 / 0.332	0.234 / 0.273
GPT4TS	0.163 / 0.215	0.210 / 0.254	0.256 / 0.292	0.321 / 0.339	0.238 / 0.275
DLinear	0.171 / 0.224	0.215 / 0.263	0.258 / 0.299	0.320 / 0.346	0.241 / 0.283
PatchTST	0.165 / 0.215	0.210 / 0.257	0.259 / 0.297	0.332 / 0.346	0.242 / 0.279
TimesNet	0.184 / 0.230	0.245 / 0.283	0.305 / 0.321	0.381 / 0.371	0.279 / 0.301
FEDformer	0.188 / 0.253	0.250 / 0.304	0.312 / 0.346	0.387 / 0.393	0.284 / 0.324
Autoformer	0.221 / 0.297	0.270 / 0.322	0.320 / 0.351	0.390 / 0.396	0.300 / 0.342
Stationary	0.192 / 0.234	0.269 / 0.295	0.370 / 0.357	0.441 / 0.405	0.318 / 0.323
ETSformer	0.199 / 0.272	0.279 / 0.332	0.356 / 0.386	0.437 / 0.448	0.318 / 0.360
LightTS	0.217 / 0.269	0.259 / 0.304	0.303 / 0.334	0.377 / 0.382	0.289 / 0.322
Informer	0.374 / 0.401	0.552 / 0.478	724 / 0.541	0.739 / 0.558	0.597 / 0.495
Reformer	0.335 / 0.380	0.522 / 0.462	0.715 / 0.535	0.611 / 0.500	0.546 / 0.469

Table S13 (continued). Electricity

Electricity					
Model	96	192	336	720	Avg
RDTU	0.141 / 0.237	0.153 / 0.247	0.171 / 0.269	0.227 / 0.318	0.173 / 0.268
Time-LLM	0.139 / 0.241	0.151 / 0.248	0.169 / 0.270	0.240 / 0.322	0.175 / 0.270
GPT4TS	0.139 / 0.237	0.156 / 0.252	0.175 / 0.270	0.233 / 0.317	0.176 / 0.269
DLinear	0.150 / 0.253	0.164 / 0.264	0.181 / 0.282	0.223 / 0.321	0.180 / 0.280
PatchTST	0.140 / 0.238	0.160 / 0.255	0.180 / 0.276	0.241 / 0.323	0.180 / 0.273
TimesNet	0.299 / 0.373	0.305 / 0.379	0.319 / 0.391	0.369 / 0.426	0.323 / 0.392
FEDformer	0.231 / 0.323	0.261 / 0.356	0.360 / 0.445	0.530 / 0.585	0.346 / 0.427
Autoformer	0.261 / 0.348	0.338 / 0.406	0.410 / 0.474	0.715 / 0.685	0.431 / 0.478
Stationary	0.420 / 0.466	0.411 / 0.459	0.434 / 0.473	0.510 / 0.521	0.444 / 0.480
ETSformer	0.599 / 0.587	0.620 / 0.598	0.662 / 0.619	0.757 / 0.664	0.660 / 0.617
LightTS	0.350 / 0.425	0.376 / 0.448	0.428 / 0.485	0.611 / 0.597	0.441 / 0.489
Informer	1.259 / 0.919	1.160 / 0.873	1.157 / 0.872	1.203 / 0.898	1.195 / 0.891
Reformer	0.993 / 0.784	0.938 / 0.753	0.925 / 0.745	1.004 / 0.790	0.965 / 0.768

Table S13 (continued). Traffic

Traffic					
Model	96	192	336	720	Avg
RDTU	0.405 / 0.288	0.412 / 0.292	0.424 / 0.306	0.463 / 0.325	0.426 / 0.303
Time-LLM	0.418 / 0.291	0.414 / 0.296	0.421 / 0.311	0.462 / 0.327	0.429 / 0.306
GPT4TS	0.414 / 0.297	0.426 / 0.301	0.434 / 0.303	0.487 / 0.337	0.440 / 0.310
DLinear	0.419 / 0.298	0.434 / 0.305	0.449 / 0.313	0.484 / 0.336	0.447 / 0.313
PatchTST	0.403 / 0.289	0.415 / 0.296	0.426 / 0.304	0.474 / 0.331	0.430 / 0.305
TimesNet	0.719 / 0.416	0.748 / 0.428	0.853 / 0.471	1.485 / 0.825	0.951 / 0.535
FEDformer	0.639 / 0.400	0.637 / 0.416	0.655 / 0.427	0.722 / 0.456	0.663 / 0.425
Autoformer	0.672 / 0.405	0.727 / 0.424	0.749 / 0.454	0.847 / 0.499	0.749 / 0.446
Stationary	1.412 / 0.802	1.419 / 0.806	1.443 / 0.815	1.539 / 0.837	1.453 / 0.815
ETSformer	1.643 / 0.855	1.641 / 0.854	1.711 / 0.878	2.660 / 1.157	1.914 / 0.936
LightTS	1.157 / 0.636	1.207 / 0.661	1.334 / 0.713	1.292 / 0.726	1.248 / 0.684
Informer	1.557 / 0.821	1.454 / 0.765	1.521 / 0.812	1.605 / 0.846	1.534 / 0.811
Reformer	1.527 / 0.815	1.538 / 0.817	1.550 / 0.819	1.588 / 0.833	1.551 / 0.821

Source-reported ranking counts. RDTU: 58, Time-LLM: 8, GPT4TS: 7, DLinear: 1, PatchTST: 1, TimesNet: 0, FEDformer: 0, Autoformer: 0, Stationary: 0, ETSformer: 0, LightTS: 0, Informer: 0, Reformer: 0. The original counting convention is not specified; these values are retained without recalculation.

Table S13 evaluates data efficiency through few-shot learning on 10% training data. In this data-scarce regime, RDTU exhibits dominant generalization capabilities, with a source-reported “1st Count” of 58, while Time-LLM has a reported count of 8. The repository source notes distinguish this reported footer from a direct count of the displayed numerical minima.

Table S14: Full few-shot forecasting results: 5% training data. Each entry is MSE / MAE; lower values are better.

ETTh1					
Model	96	192	336	720	Avg
RDTU	0.479 / 0.460	0.624 / 0.536	0.741 / 0.603	- / -	0.615 / 0.535
Time-LLM	0.483 / 0.464	0.629 / 0.540	0.768 / 0.626	- / -	0.627 / 0.543
GPT4TS	0.543 / 0.506	0.748 / 0.580	0.754 / 0.595	- / -	0.681 / 0.560
DLinear	0.547 / 0.503	0.720 / 0.604	0.984 / 0.727	- / -	0.750 / 0.611
PatchTST	0.557 / 0.519	0.711 / 0.570	0.816 / 0.619	- / -	0.694 / 0.569
TimesNet	0.892 / 0.625	0.940 / 0.665	0.945 / 0.653	- / -	0.925 / 0.647
FEDformer	0.593 / 0.529	0.652 / 0.563	0.731 / 0.594	- / -	0.658 / 0.562
Autoformer	0.681 / 0.570	0.725 / 0.602	0.761 / 0.624	- / -	0.722 / 0.598
Stationary	0.952 / 0.650	0.943 / 0.645	0.935 / 0.644	- / -	0.943 / 0.646
ETSformer	1.169 / 0.832	1.221 / 0.853	1.179 / 0.832	- / -	1.189 / 0.839
LightTS	1.483 / 0.91	1.525 / 0.93	1.347 / 0.87	- / -	1.451 / 0.903
Informer	1.225 / 0.812	1.249 / 0.828	1.202 / 0.811	- / -	1.225 / 0.817
Reformer	1.198 / 0.795	1.273 / 0.853	1.254 / 0.857	- / -	1.241 / 0.835

Table S14 (continued). ETTh2

ETTh2					
Model	96	192	336	720	Avg
RDTU	0.332 / 0.394	0.405 / 0.423	0.410 / 0.438	- / -	0.382 / 0.418
Time-LLM	0.336 / 0.397	0.406 / 0.425	0.405 / 0.432	- / -	0.382 / 0.418
GPT4TS	0.376 / 0.421	0.418 / 0.441	0.408 / 0.439	- / -	0.400 / 0.433
DLinear	0.442 / 0.456	0.617 / 0.542	1.424 / 0.849	- / -	0.694 / 0.577
PatchTST	0.401 / 0.421	0.452 / 0.455	0.464 / 0.469	- / -	0.827 / 0.615
TimesNet	0.409 / 0.420	0.483 / 0.464	0.499 / 0.479	- / -	0.439 / 0.448
FEDformer	0.390 / 0.424	0.457 / 0.465	0.477 / 0.483	- / -	0.463 / 0.454
Autoformer	0.428 / 0.468	0.496 / 0.504	0.486 / 0.496	- / -	0.441 / 0.457
Stationary	0.408 / 0.423	0.497 / 0.468	0.507 / 0.481	- / -	0.470 / 0.489
ETSformer	0.678 / 0.619	0.845 / 0.697	0.905 / 0.727	- / -	0.809 / 0.681
LightTS	2.022 / 1.006	3.534 / 1.348	4.063 / 1.451	- / -	3.206 / 1.268
Informer	3.837 / 1.508	3.975 / 1.933	3.956 / 1.520	- / -	3.922 / 1.653
Reformer	3.753 / 1.518	3.516 / 1.473	3.312 / 1.427	- / -	3.527 / 1.472

Table S14 (continued). ETTm1

ETTm1					
Model	96	192	336	720	Avg
RDTU	0.313 / 0.369	0.438 / 0.427	0.446 / 0.431	0.485 / 0.478	0.421 / 0.426
Time-LLM	0.316 / 0.377	0.450 / 0.464	0.450 / 0.424	0.483 / 0.471	0.425 / 0.434
GPT4TS	0.386 / 0.405	0.440 / 0.438	0.485 / 0.459	0.577 / 0.499	0.472 / 0.450
DLinear	0.332 / 0.374	0.358 / 0.390	0.402 / 0.416	0.511 / 0.489	0.400 / 0.417
PatchTST	0.399 / 0.414	0.441 / 0.436	0.499 / 0.467	0.767 / 0.587	0.526 / 0.476
TimesNet	0.606 / 0.518	0.681 / 0.539	0.786 / 0.597	0.796 / 0.593	0.717 / 0.561
FEDformer	0.628 / 0.544	0.666 / 0.566	0.807 / 0.628	0.822 / 0.633	0.730 / 0.592
Autoformer	0.726 / 0.578	0.750 / 0.591	0.851 / 0.659	0.857 / 0.655	0.796 / 0.620
Stationary	0.823 / 0.587	0.844 / 0.591	0.870 / 0.603	0.893 / 0.611	0.857 / 0.598
ETSformer	1.031 / 0.747	1.087 / 0.766	1.138 / 0.787	1.245 / 0.831	1.125 / 0.782
LightTS	1.048 / 0.733	1.097 / 0.756	1.147 / 0.775	1.200 / 0.799	1.123 / 0.765
Informer	1.130 / 0.775	1.150 / 0.788	1.198 / 0.809	1.175 / 0.794	1.163 / 0.791
Reformer	1.234 / 0.798	1.287 / 0.839	1.288 / 0.842	1.247 / 0.828	1.264 / 0.826

Table S14 (continued). ETTm2

ETTm2					
Model	96	192	336	720	Avg
RDTU	0.176 / 0.260	0.213 / 0.286	0.271 / 0.327	0.436 / 0.415	0.274 / 0.322
Time-LLM	0.174 / 0.261	0.215 / 0.287	0.273 / 0.330	0.433 / 0.412	0.274 / 0.323
GPT4TS	0.199 / 0.280	0.256 / 0.316	0.318 / 0.353	0.460 / 0.436	0.308 / 0.346
DLinear	0.236 / 0.326	0.306 / 0.373	0.380 / 0.423	0.674 / 0.583	0.399 / 0.426
PatchTST	0.206 / 0.288	0.264 / 0.324	0.334 / 0.367	0.454 / 0.432	0.314 / 0.352
TimesNet	0.220 / 0.299	0.311 / 0.361	0.338 / 0.366	0.509 / 0.465	0.344 / 0.372
FEDformer	0.229 / 0.320	0.394 / 0.361	0.378 / 0.427	0.523 / 0.510	0.381 / 0.404
Autoformer	0.232 / 0.322	0.291 / 0.357	0.478 / 0.517	0.553 / 0.538	0.388 / 0.433
Stationary	0.238 / 0.316	0.298 / 0.349	0.353 / 0.380	0.475 / 0.445	0.341 / 0.372
ETSformer	0.404 / 0.485	0.479 / 0.521	0.552 / 0.555	0.701 / 0.627	0.534 / 0.547
LightTS	1.108 / 0.772	1.317 / 0.850	1.415 / 0.879	1.822 / 0.984	1.415 / 0.871
Informer	3.599 / 1.478	3.578 / 1.475	3.561 / 1.473	3.896 / 1.533	3.658 / 1.489
Reformer	3.883 / 1.545	3.553 / 1.484	3.446 / 1.460	3.445 / 1.460	3.581 / 1.487

Table S14 (continued). Weather

Weather					
Model	96	192	336	720	Avg
RDTU	0.168 / 0.220	0.220 / 0.267	0.278 / 0.317	0.367 / 0.384	0.258 / 0.297
Time-LLM	0.172 / 0.263	0.224 / 0.271	0.282 / 0.321	0.366 / 0.381	0.260 / 0.309
GPT4TS	0.175 / 0.230	0.227 / 0.276	0.286 / 0.322	0.366 / 0.379	0.263 / 0.301
DLinear	0.184 / 0.242	0.228 / 0.283	0.279 / 0.322	0.364 / 0.388	0.263 / 0.308
PatchTST	0.171 / 0.224	0.230 / 0.277	0.294 / 0.326	0.384 / 0.387	0.269 / 0.303
TimesNet	0.207 / 0.253	0.272 / 0.307	0.313 / 0.328	0.400 / 0.385	0.298 / 0.318
FEDformer	0.229 / 0.309	0.265 / 0.317	0.353 / 0.392	0.391 / 0.394	0.309 / 0.353
Autoformer	0.227 / 0.299	0.278 / 0.333	0.351 / 0.393	0.387 / 0.389	0.310 / 0.353
Stationary	0.215 / 0.252	0.290 / 0.307	0.353 / 0.348	0.452 / 0.407	0.327 / 0.328
ETSformer	0.218 / 0.295	0.294 / 0.331	0.359 / 0.398	0.461 / 0.461	0.333 / 0.371
LightTS	0.230 / 0.285	0.274 / 0.323	0.318 / 0.355	0.401 / 0.418	0.305 / 0.345
Informer	0.497 / 0.497	0.620 / 0.545	0.649 / 0.547	0.570 / 0.522	0.584 / 0.527
Reformer	0.406 / 0.435	0.446 / 0.450	0.465 / 0.459	0.471 / 0.468	0.447 / 0.453

Table S14 (continued). Electricity

Electricity					
Model	96	192	336	720	Avg
RDTU	0.143 / 0.238	0.155 / 0.240	0.174 / 0.273	0.220 / 0.308	0.173 / 0.265
Time-LLM	0.147 / 0.242	0.158 / 0.241	0.178 / 0.277	0.224 / 0.312	0.179 / 0.268
GPT4TS	0.143 / 0.241	0.159 / 0.255	0.179 / 0.274	0.233 / 0.323	0.178 / 0.273
DLinear	0.150 / 0.251	0.163 / 0.263	0.175 / 0.278	0.219 / 0.311	0.176 / 0.275
PatchTST	0.145 / 0.244	0.163 / 0.260	0.183 / 0.281	0.233 / 0.323	0.181 / 0.277
TimesNet	0.315 / 0.389	0.318 / 0.396	0.340 / 0.415	0.635 / 0.613	0.402 / 0.453
FEDformer	0.235 / 0.322	0.247 / 0.341	0.267 / 0.356	0.318 / 0.394	0.266 / 0.353
Autoformer	0.297 / 0.367	0.308 / 0.375	0.354 / 0.411	0.426 / 0.466	0.346 / 0.404
Stationary	0.484 / 0.518	0.501 / 0.531	0.574 / 0.578	0.952 / 0.786	0.627 / 0.603
ETSformer	0.697 / 0.638	0.718 / 0.648	0.758 / 0.667	1.028 / 0.788	0.800 / 0.685
LightTS	0.639 / 0.609	0.772 / 0.678	0.901 / 0.745	1.200 / 0.871	0.878 / 0.725
Informer	1.265 / 0.919	1.298 / 0.939	1.302 / 0.942	1.259 / 0.919	1.281 / 0.929
Reformer	1.414 / 0.855	1.240 / 0.919	1.253 / 0.921	1.249 / 0.921	1.289 / 0.904

Table S14 (continued). Traffic

Traffic					
Model	96	192	336	720	Avg
RDTU	0.410 / 0.287	0.415 / 0.290	0.439 / 0.316	- / -	0.421 / 0.298
Time-LLM	0.414 / 0.291	0.419 / 0.291	0.437 / 0.314	- / -	0.423 / 0.298
GPT4TS	0.419 / 0.298	0.434 / 0.305	0.449 / 0.313	- / -	0.434 / 0.305
DLinear	0.427 / 0.304	0.447 / 0.315	0.478 / 0.333	- / -	0.450 / 0.317
PatchTST	0.404 / 0.286	0.412 / 0.294	0.439 / 0.310	- / -	0.418 / 0.296
TimesNet	0.854 / 0.492	0.894 / 0.517	0.853 / 0.471	- / -	0.867 / 0.493
FEDformer	0.670 / 0.421	0.653 / 0.405	0.707 / 0.445	- / -	0.676 / 0.423
Autoformer	0.795 / 0.481	0.837 / 0.503	0.867 / 0.523	- / -	0.833 / 0.502
Stationary	1.468 / 0.821	1.509 / 0.838	1.602 / 0.860	- / -	1.526 / 0.839
ETSformer	1.643 / 0.855	1.856 / 0.928	2.080 / 0.999	- / -	1.859 / 0.927
LightTS	1.157 / 0.636	1.688 / 0.848	1.826 / 0.903	- / -	1.557 / 0.795
Informer	1.557 / 0.821	1.596 / 0.834	1.621 / 0.841	- / -	1.591 / 0.832
Reformer	1.586 / 0.841	1.602 / 0.844	1.668 / 0.868	- / -	1.618 / 0.851

Source-reported ranking counts. RDTU: 39, Time-LLM: 10, GPT4TS: 2, DLinear: 8, PatchTST: 6, TimesNet: 0, FEDformer: 2, Autoformer: 0, Stationary: 0, ETSformer: 0, LightTS: 0, Informer: 0, Reformer: 0. The original counting convention is not specified; these values are retained without recalculation.

Table S14 further investigates model robustness under extreme data scarcity by reducing the training data to 5%. Despite the limited information, RDTU maintains a strong lead over baseline methods, achieving the lowest error rates in 39 instances (“1st Count”). In comparison, the closest competitor, Time-LLM, records 10 wins, while DLinear achieves 8. Although certain long-horizon forecasts (e.g., $H = 720$) could not be computed for

some datasets due to insufficient training samples, RDTU consistently outperforms peer models on the available horizons, demonstrating superior few-shot adaptability.

C.2 Zero-Shot Forecasting

Table S15: Full zero-shot ETT transfer results. Each entry is MSE / MAE; lower values are better.

ETTh1 → ETTh2					
Model	96	192	336	720	Avg
RDTU	0.272 / 0.329	0.343 / 0.368	0.378 / 0.407	0.384 / 0.413	0.344 / 0.379
Time-LLM	0.279 / 0.337	0.351 / 0.374	0.388 / 0.415	0.391 / 0.420	0.353 / 0.387
LLMTime	0.510 / 0.576	0.523 / 0.586	0.640 / 0.637	2.296 / 1.034	0.992 / 0.708
GPT4TS	0.335 / 0.374	0.412 / 0.417	0.441 / 0.444	0.438 / 0.452	0.406 / 0.422
DLinear	0.347 / 0.400	0.447 / 0.460	0.515 / 0.505	0.665 / 0.589	0.493 / 0.488
PatchTST	0.304 / 0.350	0.386 / 0.400	0.414 / 0.428	0.419 / 0.443	0.380 / 0.405
TimesNet	0.358 / 0.387	0.427 / 0.429	0.449 / 0.451	0.448 / 0.458	0.421 / 0.431
Autoformer	0.469 / 0.486	0.634 / 0.567	0.655 / 0.588	0.570 / 0.549	0.582 / 0.548

Table S15 (continued). ETTh1 → ETTm2

ETTh1 → ETTm2					
Model	96	192	336	720	Avg
RDTU	0.178 / 0.282	0.231 / 0.307	0.287 / 0.359	0.367 / 0.383	0.266 / 0.333
Time-LLM	0.189 / 0.293	0.237 / 0.312	0.291 / 0.365	0.372 / 0.390	0.273 / 0.340
LLMTime	0.646 / 0.563	0.934 / 0.654	1.157 / 0.728	4.730 / 1.531	1.867 / 0.869
GPT4TS	0.236 / 0.315	0.287 / 0.342	0.341 / 0.374	0.435 / 0.422	0.325 / 0.363
DLinear	0.255 / 0.357	0.338 / 0.413	0.425 / 0.465	0.640 / 0.573	0.415 / 0.452
PatchTST	0.215 / 0.304	0.275 / 0.339	0.334 / 0.373	0.431 / 0.424	0.314 / 0.360
TimesNet	0.239 / 0.313	0.291 / 0.342	0.342 / 0.371	0.434 / 0.419	0.327 / 0.361
Autoformer	0.352 / 0.432	0.413 / 0.460	0.465 / 0.489	0.599 / 0.551	0.457 / 0.483

Table S15 (continued). ETTh2 → ETTh1

ETTh2 → ETTh1					
Model	96	192	336	720	Avg
RDTU	0.446 / 0.449	0.462 / 0.459	0.503 / 0.480	0.509 / 0.495	0.480 / 0.471
Time-LLM	0.450 / 0.452	0.465 / 0.461	0.501 / 0.482	0.501 / 0.502	0.479 / 0.474
LLMTime	1.130 / 0.777	1.242 / 0.820	1.328 / 0.864	4.145 / 1.461	1.961 / 0.981
GPT4TS	0.732 / 0.577	0.758 / 0.559	0.759 / 0.578	0.781 / 0.597	0.757 / 0.578
DLinear	0.689 / 0.555	0.707 / 0.568	0.710 / 0.577	0.704 / 0.596	0.703 / 0.574
PatchTST	0.485 / 0.465	0.565 / 0.509	0.581 / 0.515	0.628 / 0.561	0.565 / 0.513
TimesNet	0.848 / 0.601	0.860 / 0.610	0.867 / 0.626	0.887 / 0.648	0.865 / 0.621
Autoformer	0.693 / 0.569	0.760 / 0.601	0.781 / 0.619	0.796 / 0.644	0.757 / 0.608

Table S15 (continued). ETTh2 → ETTm2

ETTh2 → ETTm2					
Model	96	192	336	720	Avg
RDTU	0.169 / 0.269	0.228 / 0.307	0.288 / 0.330	0.387 / 0.411	0.268 / 0.329
Time-LLM	0.174 / 0.276	0.233 / 0.315	0.291 / 0.337	0.392 / 0.417	0.272 / 0.341
LLMTime	0.646 / 0.563	0.934 / 0.654	1.157 / 0.728	4.730 / 1.531	1.867 / 0.869
GPT4TS	0.253 / 0.329	0.293 / 0.346	0.347 / 0.376	0.446 / 0.429	0.335 / 0.370
DLinear	0.240 / 0.336	0.295 / 0.369	0.345 / 0.397	0.432 / 0.442	0.328 / 0.386
PatchTST	0.226 / 0.309	0.289 / 0.345	0.348 / 0.379	0.439 / 0.427	0.325 / 0.365
TimesNet	0.248 / 0.324	0.296 / 0.352	0.353 / 0.383	0.471 / 0.446	0.342 / 0.376
Autoformer	0.263 / 0.352	0.326 / 0.389	0.387 / 0.426	0.487 / 0.478	0.366 / 0.411

Table S15 (continued). ETTm1 $\rightarrow$ ETTh2

ETTm1 $\rightarrow$ ETTh2					
Model	96	192	336	720	Avg
RDTU	0.319 / 0.363	0.383 / 0.406	0.401 / 0.428	0.407 / 0.433	0.378 / 0.408
Time-LLM	0.321 / 0.369	0.389 / 0.410	0.408 / 0.433	0.406 / 0.436	0.381 / 0.412
LLMTime	0.510 / 0.576	0.523 / 0.586	0.640 / 0.637	2.296 / 1.034	0.992 / 0.708
GPT4TS	0.353 / 0.392	0.443 / 0.437	0.469 / 0.461	0.466 / 0.468	0.433 / 0.439
DLinear	0.365 / 0.415	0.454 / 0.462	0.496 / 0.494	0.541 / 0.529	0.464 / 0.475
PatchTST	0.354 / 0.385	0.447 / 0.434	0.481 / 0.463	0.474 / 0.471	0.439 / 0.438
TimesNet	0.377 / 0.407	0.471 / 0.453	0.472 / 0.484	0.495 / 0.482	0.457 / 0.454
Autoformer	0.435 / 0.470	0.495 / 0.489	0.470 / 0.472	0.480 / 0.485	0.470 / 0.479

Table S15 (continued). ETTm1 $\rightarrow$ ETTm2

ETTm1 $\rightarrow$ ETTm2					
Model	96	192	336	720	Avg
RDTU	0.165 / 0.254	0.225 / 0.315	0.282 / 0.334	0.372 / 0.362	0.261 / 0.316
Time-LLM	0.169 / 0.257	0.227 / 0.318	0.290 / 0.338	0.375 / 0.367	0.268 / 0.320
LLMTime	0.646 / 0.563	0.934 / 0.654	1.157 / 0.728	4.730 / 1.531	1.867 / 0.869
GPT4TS	0.217 / 0.294	0.277 / 0.327	0.331 / 0.360	0.429 / 0.413	0.313 / 0.348
DLinear	0.221 / 0.314	0.286 / 0.359	0.357 / 0.406	0.476 / 0.476	0.335 / 0.389
PatchTST	0.195 / 0.271	0.258 / 0.311	0.317 / 0.348	0.416 / 0.404	0.296 / 0.334
TimesNet	0.222 / 0.295	0.288 / 0.337	0.341 / 0.367	0.436 / 0.418	0.322 / 0.354
Autoformer	0.385 / 0.457	0.433 / 0.469	0.476 / 0.477	0.582 / 0.535	0.469 / 0.484

Table S15 (continued). ETTm2 $\rightarrow$ ETTh2

ETTm2 $\rightarrow$ ETTh2					
Model	96	192	336	720	Avg
RDTU	0.295 / 0.351	0.356 / 0.392	0.363 / 0.407	0.388 / 0.429	0.351 / 0.395
Time-LLM	0.298 / 0.356	0.359 / 0.397	0.367 / 0.412	0.393 / 0.434	0.354 / 0.400
LLMTime	0.510 / 0.576	0.523 / 0.586	0.640 / 0.637	2.296 / 1.034	0.992 / 0.708
GPT4TS	0.360 / 0.401	0.434 / 0.437	0.460 / 0.459	0.485 / 0.477	0.435 / 0.443
DLinear	0.333 / 0.391	0.441 / 0.456	0.505 / 0.503	0.543 / 0.534	0.455 / 0.471
PatchTST	0.327 / 0.367	0.411 / 0.418	0.439 / 0.447	0.459 / 0.470	0.409 / 0.425
TimesNet	0.360 / 0.401	0.434 / 0.437	0.460 / 0.459	0.485 / 0.477	0.435 / 0.443
Autoformer	0.353 / 0.393	0.432 / 0.437	0.452 / 0.459	0.453 / 0.467	0.423 / 0.439

Table S15 (continued). ETTm2 $\rightarrow$ ETTm1

ETTm2 $\rightarrow$ ETTm1					
Model	96	192	336	720	Avg
RDTU	0.356 / 0.394	0.386 / 0.414	0.419 / 0.440	0.485 / 0.487	0.412 / 0.434
Time-LLM	0.359 / 0.397	0.390 / 0.420	0.421 / 0.445	0.487 / 0.488	0.414 / 0.438
LLMTime	1.179 / 0.781	1.327 / 0.846	1.478 / 0.902	3.749 / 1.408	1.933 / 0.984
GPT4TS	0.747 / 0.558	0.781 / 0.560	0.778 / 0.578	0.769 / 0.573	0.769 / 0.567
DLinear	0.570 / 0.490	0.590 / 0.506	0.706 / 0.567	0.731 / 0.584	0.649 / 0.537
PatchTST	0.491 / 0.437	0.530 / 0.470	0.565 / 0.497	0.686 / 0.565	0.568 / 0.492
TimesNet	0.747 / 0.558	0.781 / 0.560	0.778 / 0.578	0.769 / 0.573	0.769 / 0.567
Autoformer	0.735 / 0.576	0.753 / 0.586	0.750 / 0.593	0.782 / 0.609	0.755 / 0.591

Table S15 presents the zero-shot learning results on ETT datasets, evaluating the transferability of models between different domains (e.g., $ETT_{h1} \rightarrow ETT_{h2}$). RDTU demonstrates exceptional cross-domain generalization, achieving the best performance (highlighted in red) across the vast majority of transfer scenarios and horizons. While Time-LLM frequently secures the second-best position (blue) and PatchTST consistently ranks third (green), RDTU significantly outperforms both, particularly in challenging transfer tasks such as $ETT_{m1} \rightarrow ETT_{h2}$ and $ETT_{m2} \rightarrow ETT_{h1}$, validating the efficacy of its language-based representations for zero-shot forecasting.

C.3 Zero-shot Length Generalization

Table S16: Zero-shot length generalization of RDTU. Models trained only up to $H = 336$ under the 5% data regime are directly evaluated on $H = 720$ without additional training, and are compared with RDTU trained on $H = 720$ using 10% data. Each entry is MSE / MAE; both columns are evaluated at $H = 720$.

Dataset	Train through $H = 336$ (5%)	Train on $H = 720$ (10%)
ETTh1	0.705 / 0.598	0.694 / 0.587
ETTh2	0.442 / 0.456	0.425 / 0.443
Traffic	0.478 / 0.337	0.463 / 0.325

A core advantage of RDTU is its zero-shot length generalization ability. Unlike conventional forecasting models whose output heads are often tied to a fixed prediction horizon, RDTU formulates forecasting as an autoregressive token generation task. This formulation naturally relaxes fixed output-dimensional constraints and allows the model to generate longer horizons at inference time without modifying the architecture or conducting additional horizon-specific training.

As shown in Table S16, RDTU trained only up to $H = 336$ under the 5% data regime can be directly evaluated on $H = 720$. Although a moderate performance gap is expected due to longer-horizon error accumulation, the extrapolated model remains highly competitive compared with the model trained directly on $H = 720$ using 10% data. For example, on ETTh1, the zero-shot length extrapolation setting obtains 0.705 MSE and 0.598 MAE, which is close to the 0.694 MSE and 0.587 MAE achieved by the stronger 10% $H = 720$ training setting. Similar trends are observed on ETTh2 and Traffic, where the performance degradation remains small despite the absence of any $H = 720$ training samples. These results demonstrate that RDTU does not merely memorize a fixed output length, but learns a flexible generation policy that can extrapolate to unseen prediction horizons.

Table S17: Cross-domain zero-shot generalization results. All zero-shot models are trained on ETTh1 and directly evaluated on target datasets from different domains without target-domain training. Lower MSE and MAE indicate better performance. Each entry is MSE / MAE.

Dataset	Supervised RDTU	Zero-shot RDTU	Zero-shot Time-LLM	Zero-shot GPT4TS
Electricity	0.156 / 0.246	0.182 / 0.270	0.254 / 0.312	0.328 / 0.385
Weather	0.221 / 0.251	0.245 / 0.278	0.386 / 0.395	0.462 / 0.431
Traffic	0.389 / 0.255	0.421 / 0.295	0.652 / 0.415	0.735 / 0.484
Illness	1.559 / 0.812	1.645 / 0.910	2.642 / 1.753	3.215 / 2.058

C.4 Cross-Domain Zero-Shot Generalization

To further verify whether RDTU learns transferable temporal dynamics rather than relying on dataset-specific tuning, we conduct a more challenging cross-domain zero-shot experiment. Different from the ETT-family transfer setting in Table S4, this experiment trains the model only on ETTh1 and directly evaluates it on target datasets from substantially different domains, including Electricity, Weather, Traffic, and Illness.

As shown in Table S17, Zero-Shot RDTU consistently outperforms zero-shot Time-LLM and GPT4TS across all target datasets. For example, on Electricity, RDTU achieves 0.182 MSE and 0.270 MAE, substantially better than Time-LLM with 0.254 MSE and 0.312 MAE. On more heterogeneous domains such as Traffic and Illness, the advantage becomes even more pronounced, indicating that RDTU maintains robust forecasting ability under severe distribution shifts. Moreover, Zero-Shot RDTU remains close to the supervised RDTU upper bound, despite using no target-domain training data. These results provide stronger evidence that the proposed direct temporal unification and reinforcement-driven refinement help the model capture intrinsic and transferable time-series patterns, rather than merely fitting dataset-specific statistics or hyperparameters.

D Priority-Based Dual-Stream Hard Sample Mining

The motivation behind Dual-Stream Hard Sample Mining is driven by the STIT model’s specific failure modes and the optimization dynamics of Reinforcement Learning. After the STIT stage, the model acquires basic forecasting capabilities but suffers from critical precision deficits and formatting hallucinations. To address these, we introduce a sample selection strategy tailored specifically for RL in time series forecasting.

Algorithm S1 details the proposed **Priority-Based Dual-Stream Hard Sample Mining** strategy used to construct the RDFR dataset $\mathcal{D}_{RDFR}$. The process consists of four phases starting with initial inference and evaluation. It then bifurcates into two distinct selection streams: a *Syntactic Correction Stream* that prioritizes samples exhibiting formatting errors or low format scores (S_{fmt}) to ensure structural validity, and a *Semantic Reinforcement Stream* that targets “hard” samples with high MSE loss (L_{mse}) using a density-based binning approach to enhance prediction accuracy. Finally, the outputs of these streams are fused to create a balanced dataset for model optimization.

Algorithm S1 Priority-Based Dual-Stream Hard Sample Mining

Input: Training dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^M$, STIT Model π_θ , Quotas N_{syn}, N_{sem}

Output: RDFR Dataset $\mathcal{D}_{RDFR}$

// Phase 1: Inference & Evaluation

Initialize candidate sets $\mathcal{P}_{error} \leftarrow \emptyset, \mathcal{R}_{focus} \leftarrow \emptyset$

for $i = 1$ **to** M **do**

$\hat{y}_i \leftarrow \pi_\theta(x_i)$ // Inference

 Calculate Format Score $S_{fmt}^{(i)}$ and MSE Loss $L_{mse}^{(i)}$

if detected specific patterns in $\hat{y}_i$ (e.g., missing dot) **then**

 Add (x_i, y_i) to $\mathcal{P}_{error}$

end if

end for

// Phase 2: Syntactic Correction Stream ($\mathcal{D}_{syn}$)

if $|\mathcal{P}_{error}| > N_{syn}$ **then**

$\mathcal{D}_{syn} \leftarrow \text{Sample}(\mathcal{P}_{error}, N_{syn})$

else

$N_{remain} \leftarrow N_{syn} - |\mathcal{P}_{error}|$

$\mathcal{P}_{rank} \leftarrow \text{TopK}(\mathcal{D} \setminus \mathcal{P}_{error}, \text{key} = -S_{fmt}, k = N_{remain})$

$\mathcal{D}_{syn} \leftarrow \mathcal{P}_{error} \cup \mathcal{P}_{rank}$

end if

// Phase 3: Semantic Reinforcement Stream ($\mathcal{D}_{sem}$)

Calculate mean MSE μ_{mse} over $\mathcal{D}$

$\mathcal{R}_{focus} \leftarrow \{(x_i, y_i) \in \mathcal{D} \mid L_{mse}^{(i)} > \mu_{mse}\}$

Partition $\mathcal{R}_{focus}$ into K bins based on MSE density

$\mathcal{D}_{sem} \leftarrow \emptyset$

for $k = 1$ **to** K **do**

 Calculate quota n_k proportional to bin density

$\mathcal{D}_{sem} \leftarrow \mathcal{D}_{sem} \cup \text{Sample}(\text{Bin}_k, n_k)$

end for

// Phase 4: Fusion

$\mathcal{D}_{RDFR} \leftarrow \mathcal{D}_{syn} \cup \mathcal{D}_{sem}$

return $\mathcal{D}_{RDFR}$

We avoid full-dataset RL training during the RDFR stage for two reasons: it introduces immense computational overhead, and more importantly, it dilutes targeted reward signals. Standard "easy" samples do not trigger structural or precision failures; thus, including them prevents the RL policy from focusing on critical errors. Our preliminary observations indicate that full-dataset RL training yields inferior performance compared to training exclusively on selected hard samples (See Table S18). By actively filtering for samples that expose structural weaknesses (via the Syntactic Correction stream) and regression boundaries (via the Semantic Reinforcement stream), Dual-Stream Hard Sample Mining ensures the RDFR phase receives highly concentrated, informative reward signals. This targeted approach efficiently resolves the STIT model’s inherent flaws, directly enabling our superior forecasting performance.

To further validate the efficiency of the proposed Dual-Stream Hard Sample Mining strategy, we compare RDFR training using the full training set and the selected RDFR subset on ETTh1 with $H = 96$. As shown in Table S18, training RDFR on the full dataset substantially increases the computational cost, requiring more than 3 hours, but does not bring additional performance gains. In contrast, the proposed Dual-Stream selected subset achieves

Table S18: Ablation study on RDFR training data construction on ETTh1 with prediction horizon $H = 96$. We compare RDFR trained on the full training set and the proposed Dual-Stream selected subset.

RDFR Training Data	MSE ↓	MAE ↓	Training Time
Full Dataset	0.358	0.388	> 3h
Dual-Stream Selected (Ours)	0.351	0.382	0.3h

lower MSE and MAE while reducing the RDFR training time to only 0.3 hours. This suggests that RDFR benefits more from targeted high-leverage samples than from simply increasing the amount of reinforcement learning data, supporting the necessity of the proposed data construction strategy.

E Showcases

Figure S6 and Figure S7 presents more qualitative comparison showcases between the baseline STIT model and the proposed RDFR framework on the ETTh2 and ETTm1 dataset (input-96-predict-96). In Figure S6, the top row (a-c) illustrates critical limitations in the baseline, including incomplete sequence generation due to early stopping, low forecasting accuracy, and severe formatting errors where missing decimal points result in extreme outliers (specifically, values like 1.25 are erroneously output as 125). In contrast, the bottom row (d-f) demonstrates that RDFR effectively rectifies these issues, ensuring complete prediction lengths, significantly enhanced precision, and the elimination of formatting-induced anomalies.

In Figure S7, the top row highlights the limitations of the baseline STIT model, which suffers from incomplete sequence generation, low accuracy, and severe formatting hallucinations. Specifically, in case (c), the model generates an erroneous space after the decimal point (e.g., splitting -1.75 into -1 and 75), which causes the parser to misinterpret the fractional component as a massive integer outlier. The bottom row demonstrates that the proposed RDFR framework effectively rectifies these syntactic, semantic, and structural issues, ensuring continuous and robust time series prediction.

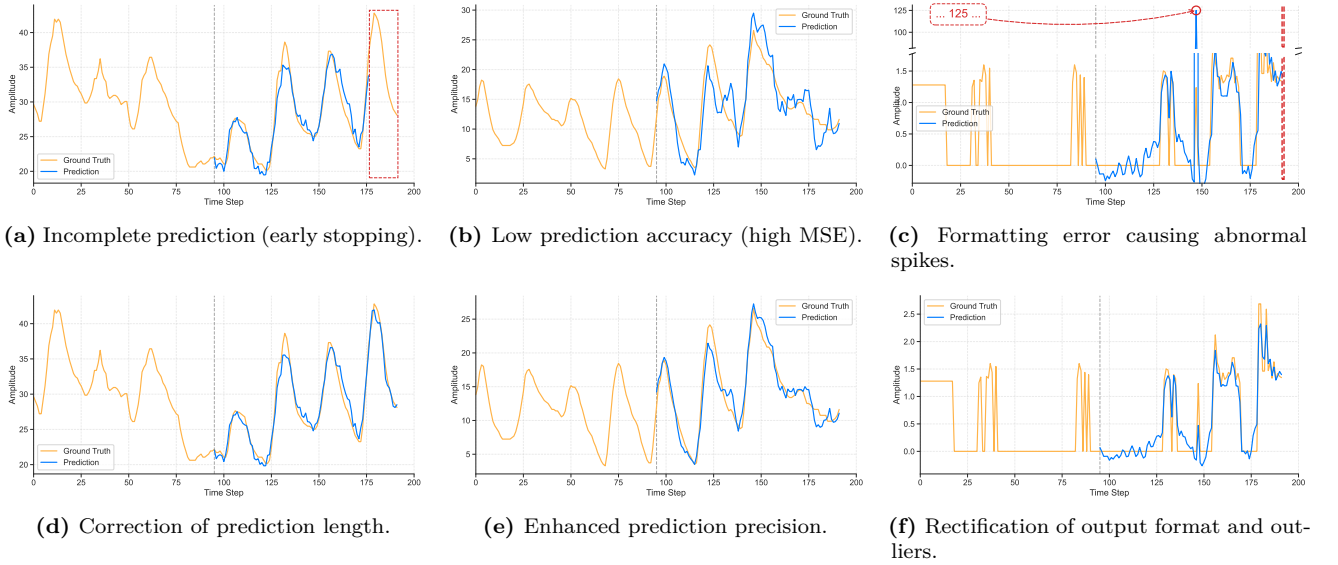


Figure S6: More showcases visualization of input-96-predict-96 results on the ETTh2 dataset. The top row (a-c) displays the baseline performance of the STIT model, which exhibits limitations in sequence length, accuracy, and output formatting. The bottom row (d-f) showcases the improvements achieved after applying the proposed RDFR framework on the corresponding samples.

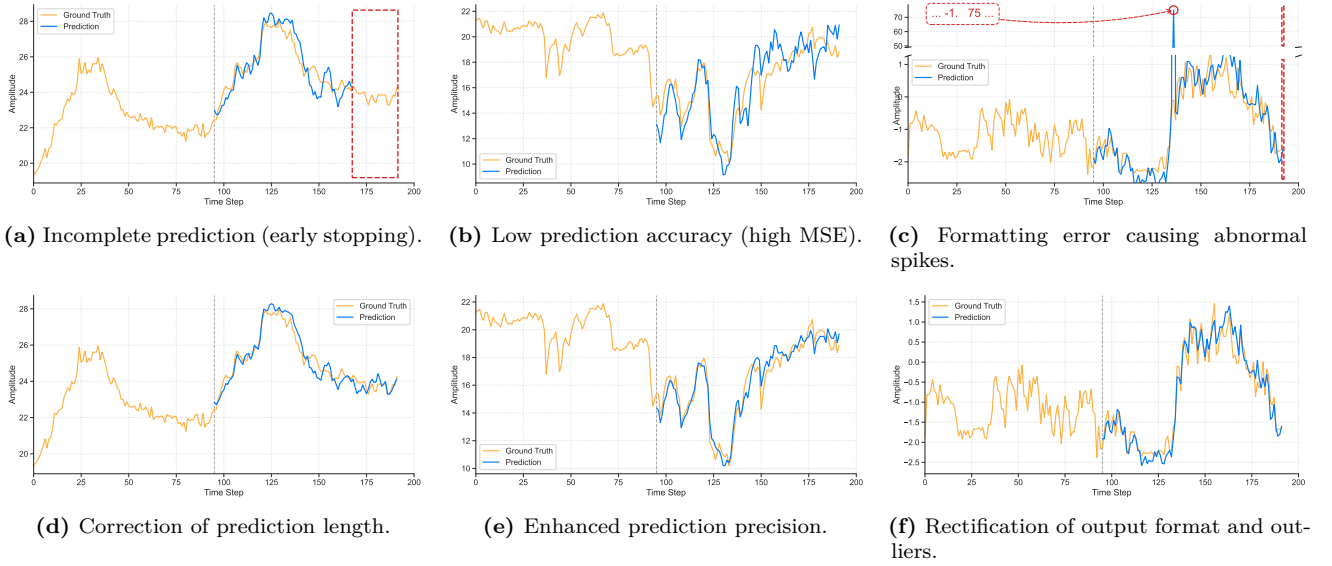


Figure S7: More showcases visualization of input-96-predict-96 results on the ETTm1 dataset. The top row (a-c) displays the baseline performance of the STIT model, which exhibits limitations in sequence length, accuracy, and output formatting. The bottom row (d-f) showcases the improvements achieved after applying the proposed RDRF framework on the corresponding samples.

F Prompt

This appendix details the specific prompt designs employed in our experiment.

Dialogue Example: Time Series Forecasting

User: You are a multivariate time series forecasting expert, and you are conducting a multivariate time series forecasting task. Variables include HUFL (High Useful Load), HULL (High Useless Load), MUFL (Middle Useful Load), MULL (Middle Useless Load), LUFL (Low Useful Load), LULL (Low Useless Load), and OT (Oil Temperature). The current time is 2016-07-09 02:00:00, and here are 96 historical data points:

2016-07-05 03:00:00, 10.717000007629396, 5.425000190734863, 8.065999984741211, 2.878000020980835, 2.8329999446868896, 1.8580000400543213, 28.913000106811523
 2016-07-05 04:00:00, 11.31999969482422, 5.625999927520752, 7.676000118255615, 2.630000114440918, 3.0460000038146973, 1.888000011444092, 30.459999084472656
 2016-07-05 05:00:00, 10.85099983215332, 4.689000129699707, 6.822999954223633, 1.883000016212464, 3.3199999332427983, 2.0409998893737797, 31.3750009442556968,

.....

2016-07-09 00:00:00, 14.60200023651123, 5.960999965667725, 11.300000190734865, 3.3399999141693115, 3.259000062942505, 1.7669999599456787, 30.952999114990234
 2016-07-09 01:00:00, 14.53499984741211, 6.162000179290772, 11.690999984741213, 3.908999919891357, 3.0460000038146973, 1.6449999809265137, 30.31999969482422
 2016-07-09 02:00:00, 14.401000022888185, 6.296000003814697, 11.72700023651123, 3.9440000057220463, 2.8020000457763667, 1.6139999628067017, 29.40500068664551
 Predict all variables for the next 96 data points.

Assistant: Here are next 96 predictions,

2016-07-09 03:00:00, 13.8761189064844679, 6.2196899796516814, 11.2944295697469368, 4.2530431436381280, 2.1881263437365770, 1.2201847471905527, 28.9887656946704766
 2016-07-09 04:00:00, 14.2560721512361734, 5.9231297870012867, 11.4640511170778296, 4.7462475899091974, 2.9999912251093641, 1.0527562380645918, 28.4748593490313695
 2016-07-09 05:00:00, 14.7053675264032115, 5.4290932353666825, 11.5004855354919862, 4.9681193018049274, 2.9424307768411118, 0.5318665943402617, 28.9633890297951275
 2016-07-09 06:00:00, 15.0472919700687022, 6.1630397546891587, 11.1093843098066891, 4.6707001161588977, 2.6000304425794396, 0.3997434904746003, 29.3292634695356469

```

2016-07-09    07:00:00,    14.2812091330150857,    6.3270099694262933,    12.3872873810183677,
4.6466697713273275, 2.3947366181772844, 0.1303528815588440, 29.4522558724904258
.....
2016-07-11    00:00:00,    16.8784257408377023,    5.0188430039810648,    5.1576691434726314,
3.9998658250940680, 3.2536048105448589, 3.6541286210987103, 30.6599944851329909
2016-07-11    01:00:00,    17.2443990451752711,    5.3244076172971182,    4.9414113159207771,
3.7657028289748924, 2.5107311848805542, 3.0150452718955765, 30.5161683922856284
2016-07-11    02:00:00,    16.6936500560171659,    5.2596094859684186,    4.8941272400060143,
2.8417151584125859, 2.2165416340952087, 2.5979149027470849, 30.9421436418460551

```

Figure S8. A representative dialogue showing the user prompt and model response.

G Societal Impacts

The proposed RDTU framework has the potential to benefit a wide range of real-world forecasting applications, such as energy management, transportation planning, finance, weather analysis, and public-resource allocation. By simplifying LLM-based time series forecasting without relying on complex pre-alignment modules, the framework may lower the engineering barrier for adapting foundation models to temporal prediction tasks and improve forecasting accessibility in data-limited scenarios. More accurate and generalizable forecasts can support better planning, reduce resource waste, and improve the reliability of decision-making systems. At the same time, as with other forecasting methods, inaccurate predictions may lead to suboptimal decisions if the model is deployed without proper validation. Therefore, practical use of the proposed framework should include domain-specific validation when applied to sensitive real-world decision-making scenarios.

Source and formatting notes

The full-manuscript source is the basis for all numerical values in this supplement. Existing summary averages, source-reported rank-count footers and statistical values are retained as supplied; the repository's source notes identify inconsistencies in those summaries. No experiments were rerun when preparing this material. The historical novelty wording was aligned with the final manuscript's narrower description of the method. The NeurIPS conference checklist author responses are archived in the repository separately from the research material; template instructions and placeholder author information are excluded.

References

- [1] Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, et al. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.
- [2] Cristian Challu, Kin G Olivares, Boris N Oreshkin, Federico Garza Ramirez, Max Mergenthaler Canseco, and Artur Dubrawski. Nhits: neural hierarchical interpolation for time series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 37, pages 6989–6997, 2023.
- [3] Mouxiang Chen, Lefei Shen, Zhuo Li, Xiaoyun Joy Wang, Jianling Sun, and Chenghao Liu. Visionts: Visual masked autoencoders are free-lunch zero-shot time series forecasters, 2024. URL <https://arxiv.org/abs/2408.17253>.
- [4] Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning*, pages 10148–10167, 2024.
- [5] DeepSeek-AI, Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruotian Ma, Mingchuan Zhang, Xiaokang Chen, Bin Xu, Wei Liu, and Chao Ruan. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning, 2025. URL <https://arxiv.org/abs/2501.12948>.
- [6] Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. Moment: A family of open time-series foundation models. *arXiv preprint arXiv:2402.03885*, 2024.
- [7] Ming Jin, Huan Yee Koh, Qingsong Wen, Daniele Zambon, Cesare Alippi, Geoffrey I. Webb, Irwin King, and Shirui Pan. A survey on graph neural networks for time series: Forecasting, classification, imputation, and anomaly detection, 2024. URL <https://arxiv.org/abs/2307.03759>.
- [8] Ming Jin, Shifan Wang, Lintao Ma, Zhixuan Chu, James Y. Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxuan Liang, Yuan-Fang Li, Shirui Pan, and Qingsong Wen. Time-llm: Time series forecasting by reprogramming large language models. In *International Conference on Learning Representations (ICLR)*, 2024.
- [9] Nikita Kitaev, Łukasz Kaiser, and Anselm Levskaya. Reformer: The efficient transformer. In *International Conference on Learning Representations*, 2020.
- [10] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, 2018. URL <https://arxiv.org/abs/1707.01926>.
- [11] Shengsheng Lin, Weiwei Lin, Xinyi Hu, Wentai Wu, Ruichao Mo, and Haocheng Zhong. Cyclenet: Enhancing time series forecasting through modeling periodic patterns. In *Thirty-eighth Conference on Neural Information Processing Systems*, 2024.
- [12] Shizhan Liu, Hang Yu, Cong Liao, Jianguo Li, Weiyao Lin, Alex X Liu, and Schahram Dustdar. Pyraformer: Low-complexity pyramidal attention for long-range time series modeling and forecasting. In *International Conference on Learning Representations*, 2022.
- [13] Xin Liu, Daniel McDuff, Geza Kovacs, Isaac Galatzer-Levy, Jacob Sunshine, Jiening Zhan, Ming-Zher Poh, Shun Liao, Paolo Di Achille, and Shwetak Patel. Large language models are few-shot health learners. *arXiv preprint arXiv:2305.15525*, 2023.
- [14] Xu Liu, Junfeng Hu, Yuan Li, Shizhe Diao, Yuxuan Liang, Bryan Hooi, and Roger Zimmermann. Unitime: A language-empowered unified model for cross-domain time series forecasting. In *Proceedings of the ACM Web Conference 2024*, 2024.
- [15] Yong Liu, Haixu Wu, Jianmin Wang, and Mingsheng Long. Non-stationary transformers: Exploring the stationarity in time series forecasting. *Advances in Neural Information Processing Systems*, 35:9881–9893, 2022.
- [16] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long. itransformer: Inverted transformers are effective for time series forecasting. *arXiv preprint arXiv:2310.06625*, 2023.
- [17] Spyros Makridakis, Evangelos Spiliotis, and Vassilios Assimakopoulos. The m4 competition: Results, findings, conclusion and way forward. *International Journal of Forecasting*, 34(4):802–808, 2018.
- [18] Shikai Qiu, Nate Gruver, Marc Finzi, and Andrew Gordon Wilson. Large Language Models Are Zero Shot Time Series Forecasters. In *Advances in Neural Information Processing Systems*, 2023.

- [19] Yuqi Nie, Nam H Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. A time series is worth 64 words: Long-term forecasting with transformers. In *International Conference on Learning Representations*, 2023.
- [20] Boris N Oreshkin, Dmitri Carpov, Nicolas Chapados, and Yoshua Bengio. N-beats: Neural basis expansion analysis for interpretable time series forecasting. In *International Conference on Learning Representations*, 2020.
- [21] ChengAo Shen, Wenchao Yu, Ziming Zhao, Dongjin Song, Wei Cheng, Haifeng Chen, and Jingchao Ni. Multi-modal view enhanced large vision models for long-term time series forecasting. In *NeurIPS*, 2025.
- [22] Artur Trindade. Electricity Load Diagrams. UCI Machine Learning Repository, 2015.
- [23] Qingsong Wen, Tian Zhou, Chaoli Zhang, Weiqi Chen, Ziqing Ma, Junchi Yan, and Liang Sun. Transformers in time series: A survey. In *Proceedings of the Thirty-Second International Joint Conference on Artificial Intelligence*, pages 6778–6786, 2022.
- [24] Gerald Woo, Chenghao Liu, Doyen Sahoo, Akshat Kumar, and Steven Hoi. Etsformer: Exponential smoothing transformers for time-series forecasting. *arXiv preprint arXiv:2202.01381*, 2022.
- [25] Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. *arXiv preprint arXiv:2402.02592*, 2024.
- [26] Haixu Wu, Jiehui Xu, Jianmin Wang, and Mingsheng Long. Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting. *Advances in Neural Information Processing Systems*, 34: 22419–22430, 2021.
- [27] Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long. Timesnet: Temporal 2d-variation modeling for general time series analysis. In *International Conference on Learning Representations*, 2023.
- [28] An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [29] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu. Are transformers effective for time series forecasting? In *Proceedings of the AAAI conference on artificial intelligence*, volume 37, pages 11121–11128, 2023.
- [30] Siru Zhong, Weilin Ruan, Ming Jin, Huan Li, Qingsong Wen, and Yuxuan Liang. Time-vlm: Exploring multimodal vision-language models for augmented time series forecasting. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025.
- [31] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, and Wancai Zhang. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 11106–11115, 2021.
- [32] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, and Rong Jin. Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting. In *International Conference on Machine Learning*, pages 27268–27286. PMLR, 2022.
- [33] Tian Zhou, Peisong Niu, Xue Wang, Liang Sun, and Rong Jin. One fits all: Power general time series analysis by pretrained lm. *Advances in Neural Information Processing Systems*, 36, 2023.